



Optimal quality oversight in kidney transplantation and its impact on transplant centers' waitlist management

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Abstract

This paper studies the effects of quality oversight in the context of assessing kidney transplantation-related outcomes and possible unintended consequences (e.g., cherry-picking of organs and selection of healthier transplant candidates). In this context, we propose a stochastic economic model that identifies socially optimal kidney transplant choices given the inherent trade-off between the expected wait time and the quality of the received donor kidney for a given patient. Socially optimal decisions seek to maximize the utilitarian welfare function defined as the sum of all patients' post-transplant expected utilities. To determine the social loss, we compare the socially optimal decisions to those taken by a transplant program that maximizes its utility. We derive the optimal quality oversight policy that minimizes social loss and examine how decisions are impacted due to the changes introduced by the new Kidney Allocation System. Our empirical analysis using data from the Scientific Registry of Transplant Recipients and **United States Renal Data System** indicates that current quality oversight imposed through Conditions of Participation results in inefficient transplant decisions for 56% of recipients, and the performance is inconsistent across different regions and parameters. We propose that the risk-adjusted post-transplant performance assessment policy considers the factors impacting demand-supply parameters such as organ availability in the 11 US transplant regions, candidates' blood type, and the newly introduced Kidney Allocation System. Policymakers and providers can utilize insights from our findings to design effective oversight mechanisms and make informed decisions regarding transplant and waitlist management that yield desired outcomes.

Keywords Kidney transplantation · Quality oversight · Queuing model in healthcare · Stochastic model · New kidney allocation system · Operations research · Operations management

Highlights

- We develop a stochastic economic model to examine how quality oversight for kidney transplant programs, focusing on performance assessment and penalizing poorly performing programs, affects various aspects of transplant outcomes. Our model establishes a citation policy that unifies changes in allocation rules, waitlist management decisions (such as accepting patients onto waitlists), transplant decisions, and quality oversight, all within a coherent framework.
- The current quality oversight policy relies solely on transplant outcomes and neglects the utilization of allocation rules. Taking into account the variation in organ availability across 11 regions and different blood types could result in a fairer citation policy and mitigate unintended consequences, such as the cherry-picking of organs.
- Based on estimations using real clinical data, we find that the current quality oversight mechanisms can be improved. For example, making adjustments to the policy based on regional organ availability and candidates' blood types could enhance the effectiveness of citation policies. Specifically, this approach has the potential to improve transplant access for higher-risk patients.
- Our findings can assist policymakers in designing effective oversight mechanisms, and transplant centers can utilize them in their decision-making processes regarding transplant and waitlist management. For instance, they can determine optimal timing for transplantation or make informed choices about which patients to accept to the center's waitlist.

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1 Introduction

Establishing a basic level of quality for services is an essential goal in a variety of sectors. For example, low quality is a significant cause for concern in healthcare, and therefore monitoring the delivered quality is imperative. To assure quality, decision-makers in healthcare have focused on *report cards* [1, 2], which are tools for collecting and publicly reporting providers' performance through a certified source [3]. The primary objective of such reporting is to make quality information available to stakeholders to support their decision-making process about appropriate medical care.

Public reporting of performance through report cards can benefit a number of important decisions in healthcare. Private organizations and government regulators can use report cards to identify underperforming providers and take action to promote higher-quality care. For example, decisions concerning the public funding of transplant programs can be based on information published through report cards. Patients and insurance companies can use report cards to search for and select providers who have better health outcomes. Such effects of report cards might lead to quality improvement because providers enhance their performance to remain competitive. Also, providers may be concerned with their reputation among peers as well as the patient population. Therefore, report cards can incentivize healthcare providers to focus on high-quality care delivery [4].

Despite the deemed benefits of report cards in improving various decisions and care quality, there are some concerns expressed about public performance reporting via report cards. In the context of this paper, two groups of concerns are relevant (see [5] for a detailed discussion of the effects of report cards in the context of mortality rates associated with coronary artery bypass surgery), which we explain next.

The first group is concerned about whether an assessment of the quality of care can be applied to healthcare services given the available data. One argument is that performance metrics derived from administrative data (i.e., mortality rates) may not be valid for managing and comparing performance outcomes across different healthcare organizations such as hospitals. Administrative data typically do not include sufficiently detailed clinical characteristics of patients, which ultimately are important determinants for outcomes [6]. An additional argument is that instead of focusing on the measurement of outcomes, the emphasis should be on process measurement. Proponents of this argument suggest that the performance of healthcare organizations should be evaluated based on what is done to improve care quality rather than measuring patient outcomes such as mortality rates. Hence, performance assessment should not be conducted independently of assessing the actual processes for providing healthcare services.

The second group is concerned about the potential unintended consequences of report card-based performance assessment and related oversight mechanisms [7, 8]. When there is substantial excess demand (a common occurrence in healthcare), outcome-based performance assessment and oversight mechanisms may not always be effective in inducing the desired behavior. An underperforming organization subject to assessment may choose not to improve its actual performance if gaming the system is less costly than actually improving the quality of care. For example, an organization may enhance its measured performance by restricting access to care for patients who are believed to carry a higher risk of negatively impacting the outcome metrics used [5, 9]. Hence, the design of report cards and the associated oversight tools (henceforth, we refer to them as *quality oversight* or just *oversight*) should account for possible unintended or even detrimental responses from providers, such as limiting access to care for high-risk patients. In relation to this concern, our research focuses on the effects of quality oversight based on publicly available report cards in measuring and assessing kidney transplantation-related outcomes and possible unintended consequences.

1.1 Report cards for kidney transplantation

Kidney transplantation is a relevant application setting for the use of report cards for quality oversight in healthcare. It is unique within the transplantation context because (i) concerns were raised about measuring the performance of transplant centers, and (ii) kidney transplant decisions are prone to gaming by providers [10]. For example, in 2007, the Centers for Medicare and Medicaid Services (CMS) implemented a policy that imposed penalties on transplant centers with poor post-transplant outcomes. The policy subjected transplant centers to a review process when post-transplant performance was below expectation and entailed citing or even shutting the centers down when the review identified systematic deficiencies [11, 12]. However, the policy had not suggested any action regarding the mortality rates of waitlisted patients. For this reason, transplant centers might not have worried about their pre-transplant outcomes (i.e., while patients are on the waitlist) as much as they were concerned with their post-transplant outcomes. Similarly, a *New York Times* article [13] mentioned that “in interviews, dozens of transplant specialists said the threat of government penalties had made doctors far more selective about the organs and patients they accepted, leading to more discards.” In addition, the proposed performance assessment was criticized for being outcome-based and not accounting for processes that could have led to transplant-related decisions. For instance, the proposed methods did not sufficiently account for the underlying donor or candidate health

factors or allocation processes [14]. Candidates waitlisted for kidney transplantation can live on dialysis, and thus centers can delay transplanting riskier patients to improve their performance measures. Transplant centers must perform a few kidney transplants per year to receive federal funding from CMS. Since most kidney transplant centers can easily meet this criterion, they can improve their outcomes by cherry-picking donor kidneys and by refusing those kidneys that are believed to pose a threat to their performance evaluation, even if the transplantation might be the superior option for the candidate [15, 16].

Despite some changes over the years, regulatory quality oversight through the requirement to disclose, assess, and manage quality persists (e.g., see CMS's Condition of Participation (CoPs) regulatory policy mentioned in §482.82).¹ Our research is motivated by past and present quality oversight policies mandated by two federal agencies, CMS and OPTN, to assess the performance of kidney transplant programs. These policies have subjected underperforming transplant centers to a quality review and improvement process, institutionalizing direct adverse financial impact. The policy has also contained provisions for excluding centers from having access to transplant organs.

1.2 Research question and contributions

The primary research question we are addressing is how to create an efficient incentive system to assess and possibly enhance the utilization of citation-based oversight mechanisms. Achieving this objective involves taking into account the dynamics of waiting lists that influence the outcomes (such as factors related to both candidates and donors that affect organ allocation and transplant decisions), and aiming to mitigate the unintended consequences of quality oversight on healthcare providers' actions. Our proposed oversight mechanism does not fully reflect the exact metrics and criteria used in current CMS and OPTN policies, as these are subject to change. Instead, we examine the penalizing nature of these

policies to understand how an optimal citation policy should be evaluated to incentivize transplant programs to improve their outcomes.² Unlike prior work, our research considers transplant-related decisions that determine outcomes and identifies factors that risk adjustment models should account for beyond clinical and medical factors of donors and recipients. This approach aims to potentially counter undesirable responses to quality oversight, such as organ cherry-picking and selective patient acceptance.

To address this question, we develop a stochastic economic model that captures how the candidate's health, organ quality, and expected wait time—shaped by the supply of and demand for kidneys—affect the utilities of the candidate and the transplant center.³ Candidates rely on service providers for consultation and recommendations during decision-making because they lack medical expertise and require the provider's cooperation to proceed with transplantation.

The model identifies the optimal decision strategies for acceptable kidney quality from three perspectives: the candidate, the service provider, and the social planner. Each stakeholder uses a threshold-based rule to decide the minimum kidney quality that aligns with their specific goals. The candidate evaluates the trade-off between the benefits of receiving a higher-quality organ and the risks of waiting too long. By factoring in organ availability and individual risk tolerance, the candidate selects a threshold to maximize their post-transplant expected utility. The service provider establishes a threshold based on its own incentives, balancing the desire to perform more transplants with a commitment to improving patient outcomes. The social planner sets a system-wide threshold designed to maximize overall welfare. They also enforce oversight measures—such as evaluating providers and implementing penalties—to align providers' actions with public interests. A comparison of these strategies shows that transplant centers may pursue incentives that conflict with socially optimal decisions. To address this misalignment, our model develops optimal incentive mechanisms, including quality oversight policies, to minimize social loss and align outcomes with the utilitarian welfare goals established by organizations such as CMS and OPTN.

¹ After receiving numerous complaints arguing for an oversight-free policy for organ transplant programs, in 2019, CMS finally reformed CoPs. According to the new policy [17], organ transplant programs were no longer obligated to submit their data to obtain Medicare reapproval. However, the previous requirements for the initial contract with CMS to enter the Medicare program and periodically disclose the transplant program's outcomes to the general public remain unchanged. In addition, CMS continues reviewing the Quality Assessment and Performance Improvement program. CMS also performs thorough investigations for public or non-public reports regarding outcomes or adverse events. Besides CMS, the Organ Procurement and Transplant Network (OPTN) also oversees transplant program outcomes. As noted by [18], to participate in organ transplantations, the transplant programs are still mandated to comply with OPTN requirements.

² In practice, a declined kidney is often offered sequentially to other waitlisted recipients until it is ultimately accepted or deemed unusable. Analyzing every step in that sequence would require a more comprehensive modeling approach, which goes beyond the scope of our current study. Here, we focus on a single acceptance decision to highlight how oversight policies may shape the choices of key participants in the transplant process. This approach captures the core elements of these policies' impact without introducing the complexities inherent in multiple reoffers.

³ In this paper, we refer to waitlisted patients seeking a kidney transplant as "candidates," as they are potential recipients of a donor organ.

The results show that the optimal quality oversight policies have to be adjusted not only for the risk due to each candidate's health and other demand-related factors but also for supply since all these factors affect a candidate's wait time and optimal kidney acceptance decision. The dependence on supply suggests that quality oversight in kidney transplantation needs to account for related factors such as region and blood type. The past and present oversight policies account for recipient and donor risk factors but do not directly adjust for supply factors. Further, the effect of supply on the optimal incentive structure makes it necessary to investigate the Kidney Allocation System (KAS)—the set of prioritization rules introduced by the United Network of Organ Sharing (UNOS) that govern which organ is offered to which candidate. Changes made to the KAS in recent years have shifted the kidney supply in favor of specific groups of healthier candidates. Yet, no change has been introduced into assessment metrics to align the incentives of involved parties.

We show that leaving the old incentive structure in place leads to sub-optimal results and characterize how the incentive structure needs to be adjusted. Using the real dataset obtained from the Scientific Registry for Transplant Recipients (SRTR) and United States Renal Data System (USRDS), we empirically illustrate our findings and quantify the impact of quality oversight on provider decisions (i.e., the minimum acceptable quality of a kidney for a given patient) and the resulting social welfare outcomes (i.e., utility based on patient survival).

Our study contributes to the understanding of quality oversight through report cards—a regulatory mechanism important for healthcare providers and policymakers. Prior operations management research examined the interrelationship between care processes and quality outcomes [e.g., see 19] and how oversight mechanisms impact outcomes [e.g., see 20, for the Hospital Readmission Reduction Program's outcomes]. Prior research also studied how to incentivize providers to ensure desirable quality performance [e.g., see 21, 22]. Our study is unique in linking these streams of research on the care processes (e.g., allocation system and resulting transplant decisions that lead to outcomes), the impact of quality oversight on outcomes, and how to design oversight mechanisms in the report card-based quality oversight context. That is, our study provides a model to study the optimal oversight mechanism that identifies trade-offs among factors relating to supply, demand, and incentives in a real-life context: kidney transplantation.

Our proposed oversight mechanism does not fully reflect the exact metrics and criteria used in current CMS and OPTN policies, as these are subject to change. Instead, we examine the penalizing nature of these policies to understand how an optimal citation policy should be evaluated to incentivize transplant programs to improve their outcomes.

Unlike prior work, our research considers transplant-related decisions that determine outcomes and identifies factors that risk adjustment models should account for, beyond clinical and medical factors of donors and recipients. This approach aims to counter potentially undesirable responses to quality oversight, such as organ cherry-picking and selective patient acceptance. In doing so, we create a holistic modeling view of transplant decisions made while candidates are on the waitlist. We also propose practical admission policies to accept candidates to the waitlist using aggregated data. In addition, our research identifies how the dependence on the supply of organs should be tied to overall kidney allocation, transplant decisions, and resulting performance assessment. Our findings can be employed by policymakers in designing effective oversight mechanisms and also by providers (e.g., transplant centers) in their transplant and waitlist management decisions (e.g., when to transplant or which patients to accept into the center's waitlist).

The rest of this paper is organized as follows. In §2, we review the relevant medical and analytical literature. The proposed model is described in §3. In §4, we derive the optimal organ acceptance decisions for both candidates and providers. In §5, we discuss optimal quality oversight and the main driving factors. In §6, we analyze the new kidney allocation system and its impact on optimal transplant decisions. In §7, we propose an admission policy for the provider to accept a candidate to its waitlist. An empirical analysis in §8 illustrates how the optimal quality oversight policy can be applied to the U.S. We discuss our findings and conclude in §9.

2 Literature review

This study contributes to two distinct subfields within the healthcare management science literature related to organ transplantation. The first subfield focuses on the clinical aspects of organ transplantation, examining the potential unintended consequences of penalty-based regulatory oversight policies on deceased donor kidney transplant outcomes. The second subfield explores policy design and decision-making processes in organ transplantation using analytical approaches. Consistent with this categorization, we first review the related works in the clinical literature, followed by an explanation of our contributions to this field. Next, we review studies in the analytical literature while highlighting our contributions to that area.

2.1 Clinical literature

The introduction of CoPs by CMS sparked significant debate within the transplant community. From a clinical perspective, there is a concern that these regulations might

reduce transplant opportunities for high-risk patients and lead to unnecessary discards of marginal-quality donor organs, potentially harming patients with limited survival chances on the waitlist. In the following discussion, we summarize the findings from the clinical literature centered on decreased transplantation volumes, increased wait times, and a risk-adjusted model that fails to capture all risks involved.

Early research by [23] reveals a significant association between low-performing centers (in this section, we refer to the level of performance from the perspective of satisfying regulatory oversight criteria) and reduced kidney transplant volume under CMS's CoP. Specifically, the study reports an average decline of 22.4 cases among low-performing centers compared to an increase of 7.8 cases in other centers. Following up this study, [15] examines the broader effects of CoPs on transplant volumes and recipient outcomes, noting a significant decrease in overall transplant volumes and donor organ utilization among non-compliant programs, with some centers experiencing declines exceeding 35%. As such, [24] finds that failure to meet CMS criteria leads to increased expected waiting times for candidates, while [25] shows that low-performance centers have higher rates of waitlist removal, further emphasizing the consequences of regulatory oversight.

Aside from the impact on volume and waitlist outcomes, a group of work within the clinical literature examines transplant decisions by centers and related outcomes as well as metrics used by the CMS. Woodside and Sung [26] examines the CMS performance evaluation program, noting that while CoPs improve transplant survival outcomes, they also increase risk-averse behavior among transplant centers, reducing resource utilization and discouraging innovation. Pelletier and Henry [27] criticizes the high sensitivity of performance assessments for leading to a high false-positive rate (falsely citing a center for not meeting the CoP criteria). Hence [28] argues the need for a comprehensive measure for evaluating transplant center performance and access to transplants, particularly the necessity of new metrics for better patient decision-making. Consistent with this argument, [29–32] criticize the risk-adjusted models in transplant center performance assessments for failing to capture all underlying clinical risk factors adequately. Beyond these clinical limitations, regional variations in organ access significantly influence transplant decisions and outcomes. For instance, [33] reveals significant geographic trends in kidney utilization, highlighting strong associations between donor kidney imports and utilization across centers. In addition to these inherent flaws, changes in organ allocation policies, such as the 2014 allocation policy change, further complicate the assessment process by affecting access to transplants. Current risk-adjusted models overlook these external factors,

potentially leading to unfair and inaccurate evaluations of transplant center performance.

Our contribution lies in addressing the unintended consequences raised in the clinical literature. Specifically, we show that risk adjustments should consider not only clinical factors but also external ones such as organ supply and patient access to transplants. By incorporating these elements, we propose a more comprehensive approach to performance assessment. Our approach emphasizes the need to reflect policy changes affecting organ supply and patient access to transplants in the models used by CMS and OPTN, ensuring that evaluations remain fair and comprehensive.

2.2 Analytical literature

Researchers in the management science community utilized analytical approaches to examine various policy aspects of organ transplantation. Related work span in general pre-transplant and transplant stages. At the pre-transplant stage, one key area of interest is the development of strategies for managing the transplant waitlist and the criteria for patient acceptance and removal. For example, [34] analyzes waitlist management for liver transplant patients using a game theory model, identifying strategies to enhance patient cooperation and improve policy recommendations. Another example study is by [35] who studies optimal screening strategies for kidney transplant patients, aiming to reduce both screening expenses and the risks of transplanting organs to unsuitable candidates. At the transplant stage, the related research, in general, seeks improvements to transplant outcomes, system efficiency, fairness, and organ utilization based on policy design using various operations research tools, including optimization, queuing theory, probabilistic methods, and mechanism design [36–38]. Extant literature broadly falls into two categories: those that study micro-level policy decisions (e.g., effects of patient choice on kidney discard rates) and those that focus on macro-level policy aspects (e.g., the impact of alternative kidney allocation policies on outcomes). At the micro level, prior research has examined how decisions such as organ acceptance and the implementation of kidney exchange programs are modeled and implemented, with a focus on enhancing efficiency [39–42]. At the macro-level, studies in organ allocation policy address broader systemic issues. For example, [43] proposes a model to reduce geographical disparities in kidney transplants by considering patient multiple-listing and providing private aviation services to wait-listed patients. Also, [44] introduces a data-driven model to evaluate the effect of UNOS proposals to modify the deceased-donor kidney allocation protocol.

Healthcare management science research specifically on oversight policies is relatively recent and sparse. Three

studies are closely related to our work. First, [45] investigates whether the oversight based on CoP influences patient selection via waiting list registrations at transplant centers. Based on structural estimation, the study finds evidence for patient selection as a result of CoP. Namely, the study finds that more aggressive centers (centers that are more aggressive with their organ utilization policies) subject to the regulation become risk-averse after CoP, leading patients to register at better-performing centers (centers that are less aggressive with their organ utilization policies), potentially generating longer waiting times for these patients.

Second and concurrent with our research is a recent study by [46] that studies how a transplant program responds to CoP policy. Focusing on regulatory oversight for lung transplant programs, they use a chance-constrained mixed-integer programming model and demonstrate that programs might limit transplants to avoid penalties, concluding that regulations can lead to more conservative patient selection and reduced transplant volumes. We differ as we develop a stylistic queuing model that retains tractability and key aspects regarding quality oversight policies for kidney transplants, waitlist management, and patient utilities. While there are similarities in the oversight policies for lung and kidney transplants, the specific criteria, monitoring protocols, and management of the waitlist differ significantly due to the unique challenges and characteristics of each type of transplant.

Third, the study by [47] takes a deeper look into how transplant centers adapt to CoP regulations. Observing that centers tend to adopt a risk-averse strategy in their patient and organ selection, it uses a multi-class queuing model based on Lyapunov functions and fluid approximation to analyze waiting list dynamics. The findings advocate for a re-evaluation of assessment metrics, suggesting that clustering transplant centers into homogeneous groups and applying cluster-based metrics could lead to more efficient and targeted outcomes.

Our study contributes to the analytical management science literature on transplant policies, specifically for regulatory oversight of kidney transplants. It does so by exploring the impact of oversight policy on kidney transplantation from a holistic perspective, integrating pre-transplant and transplant stages into a single model. Our stylistic model allows us to examine waitlist management, transplant decisions, and the impact of performance assessment from multiple stakeholder perspectives, including providers and social planners. Hence, our model captures micro-level decisions and their macro-level implications for policy in a single framework. Using our model, we evaluate the effect of penalty-based performance assessment policies, identifying key factors for designing citation policies that ensure fairness and promote desirable outcomes. Unlike previous research that typically focuses on specific aspects such as patient selection, organ acceptance, or assessment policy, our framework integrates

these decisions by considering organ supply, wait time, patient health status, patient survival, and the provider's degree of benevolence. Our proposed models offer closed-form solutions that facilitate clear understanding and strategy formulation, providing insights for designing more effective and equitable oversight policies and ultimately enhancing the efficiency and fairness of the organ allocation system.

3 Problem formulation and model description

To study the effects of quality oversight based on report cards in the context of measuring and assessment of kidney transplantation-related outcomes, we first define a basic modeling framework. This framework uses a queuing model to represent dynamics in the waitlist and defines the candidate's expected utility as a trade-off between waiting time, including the associated health deterioration, and the quality of the received kidney. The framework will be used in Sections 4 and 5 to derive a closed-form solution for optimal kidney acceptance decisions, creating an optimal quality oversight policy and proposing patient admission policy to the waitlist. The resulting closed-form solutions allow us to reason about the system's basic mechanics and show that the model can capture many essential aspects observed in the real world.

3.1 Modeling of the waitlist

To model the waitlist, we propose a tractable stylistic model for a single queue of a service provider. The model enables us to study the impact of quality oversight policy on organ acceptance strategy from the perspectives of different stakeholders. We do not model the competition among service providers nor the candidate multiple-listing options.⁴ The queue captures specific types of candidates with their matching donor kidneys based on patient-kidney matching criteria (e.g., blood type). We illustrate the corresponding queue in Fig. 1.

In the queuing setup, patients in need of kidney transplants are enlisted with the arrival rate of λ . Matching kidneys become available to the candidates in the queue according to a homogeneous Poisson process with rate μ . The assumption that kidneys arrive at the waitlist according to a Poisson process is reasonable and commonly used in the literature (e.g., see [38]). The quality of each kidney q is in $[0,1]$ and modeled as the realization from a random variable Q with

⁴ According to the literature, less than 10% of the waitlisted patients are multiple listed, and the remaining 90% are listed in a single transplant center.

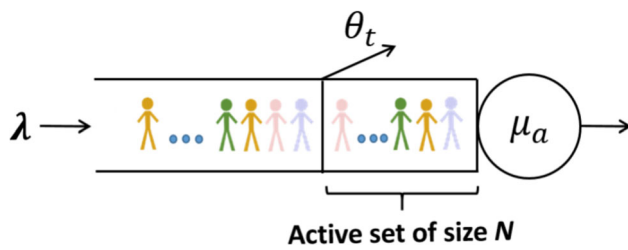


Fig. 1 The waitlist queuing model in our proposed system. Patients join the queue with rate λ . Matching kidneys arrive at the queue according to a homogeneous Poisson process with rate μ that will be offered to the candidates in the active set. The active candidates leave the queue with transplant rate μ_a . Also, candidates leave the queue with rate θ_t without receiving service due to death on the waitlist or being delisted

the distribution F_Q . Let $\mu_a \leq \mu$ be the transplant rate (service rate) of the queue, which depends on the assignment policy of the available kidneys to candidates and the candidates' choices. To reflect the fact that the arrival rate of patients joining a waitlist/queue is far larger than the rate at which donor kidneys become available, we assume λ is much larger than μ_a . The queue, however, stabilizes since candidates also leave the queue with some rate θ_t without service due to death on the waitlist or being delisted. θ_t depends on many factors and hence varies over time.

Given the scarcity of donor kidneys and the short amount of time available to perform the transplantation after extraction, only active patients⁵ at the top of the list have the chance of getting an offer. We call the set of the candidates at the top of the list the active set A (see Fig. 1). Let $|A| = N$ be the size of the active set, i.e. the number of candidates with active status. The active set size typically depends on the size of Organ Procurement Organizations (OPOs)⁶ and other factors. In kidney transplant settings, N is much smaller than the average queue size. Once a patient makes it to the active set, we call the patient a candidate.

3.2 Modeling the incentives

To focus on modeling candidates' choices, United Network for Organ Sharing (UNOS) assignment policy, and the impact of quality oversight policy on system performance metrics, we consider candidates in the active set A and model utilities and decisions of the three participants in the systems: can-

didate, service provider, and social planner (e.g., CMS and UNOS). For a patient i in the active set, let h_i be the candidate's health upon joining the active set. We define $\vec{h} = (h_i)$ for $i = 1, \dots, N$ as the health vector of the candidates in the active set, evaluated when they enter the set. We mainly focus on a single waiting list of a single service provider, and we assume the candidate's health information is public knowledge to the candidate, service provider, and social planner.

To measure the donor kidney quality in a single dimension and at an aggregated level, UNOS utilizes a linear measure called the Kidney Donor Profile Index (KDPI). The index combines multiple clinical and demographic factors of donors to translate them into one score representing kidney quality. The candidates can inform UNOS regarding their threshold on KDPI. The threshold-based decision-making model is often tractable and interpretable and sufficiently explains sophisticated collective and individual decision-making [48]. As shown later in Eq. 4, the candidate utility function increases as the kidney quality increases in our model while keeping other parameters unchanged. Therefore, to make a kidney transplant decision from the candidate's perspective, we focus on a single dimension: the quality of the offered kidney.

Assumption 3.1 We assume that decision-makers follow a threshold policy, whereby an offered kidney is accepted if its quality meets or exceeds a predefined threshold for the candidate. Specifically, we define $q \geq k_{x,i}$ where $k_{x,i}$ represents the quality threshold used by the social planner ($x = s$), the service provider ($x = p$), or the candidate ($x = c$), respectively, for candidate i , with $x \in \{s, p, c\}$.

The organ allocation and acceptance policy is complex and involves different parties with different goals, objectives, and medical considerations. In our research, we conduct our analysis at the aggregated level since our study intends to examine the potential for conflict of interest among organ transplant stockholders and identify a mechanism to realign their incentives. For these reasons, the organ allocation policy is exogenous and public in our model. Next, we present each party's objectives, information, and possible constraints:

Social planner (CMS or UNOS) The social planner decides on the performance oversight policies and considers the organ assignment. The organ assignment policy results in a probability vector $\vec{p} = (p_i)$ for $i = 1, \dots, N$ where each candidate i in the active set is offered the next available kidney with probability p_i . Since candidates cannot directly influence the assignment policy, we can ignore potential competitive behavior by the candidates. Our model assumes a fixed probability of a patient receiving an offer during the waiting period. This assumption aligns with both early transplant

⁵ Based on our interview with transplant experts, a waitlisted patient is active 80% of the time on average and inactive in the remaining time due to a variety of reasons related to health issues, insurance coverage, social factors, and so on. Regardless of patient status, inactive patients are still on the waitlist and accumulate waiting points.

⁶ OPOs are non-profit organizations tasked with recovering organs from deceased donors for transplantation in the U.S. They facilitate the testing and the delivery of recovered organs to transplant hospitals.

research by [38] and more recently by [43, 49]’s who also make similar assumptions for tractability.⁷ The fixed offer probability assumption implies that there is an expected probability of an offer a patient experiences through their waiting time. However, the patient’s total expected wait time and the corresponding utility from transplantation are influenced by the organ arrival rate, the allocation policy, and the acceptance threshold, all of which collectively determine the offer probability. Our numerical analysis in Appendix D.1 indicates that variations in organ offer probabilities slightly influence kidney acceptance decision threshold and its corresponding post-transplant expected utility, with the most substantial impact observed at lower probabilities and for sicker patients.

The information influencing the social planner’s assignment policy is the health vector $\vec{h}$ and the time each candidate has already waited for an organ. The goal of the social planner could be a set of objective functions (e.g., total social welfare and fairness). Different objective functions can be aggregated in some functional form, such as a convex combination, or the policy can be the result of multi-objective optimization. Together with the assignment policy, the social planner can determine the quality threshold for each candidate, $k_{s;i}$, that maximizes its objective function.

Service provider (transplant centers) The service provider participates in the organ acceptance decision for the candidates enrolled in their transplant center. The provider can set a minimum acceptable kidney quality threshold $k_{p;i}$. By lowering the threshold for all its candidates, the service provider could increase the number of performed transplantations while increasing the threshold would potentially improve post-transplant outcomes.

Active candidates Given the assignment policy, her health condition, and the expected wait time to receive a donor kidney offer that meets the minimum acceptable quality, the candidate and her surgeon will choose the minimum threshold for kidney quality, $k_{c;i}$, to maximize the candidate’s utility.

The timeline of events for a patient (candidate) is summarized in Fig. 2. It starts with the acceptance of the patient by a transplant center waiting to join the active set and then evaluating and rejecting offers till an acceptable kidney is offered. After the transplant, we consider one-year post-transplant outcomes to affect performance measures used for the performance oversight policy.

The total social welfare is defined as the sum of the utilities of the participants. For a given assignment policy, If the social planner maximizes the sum of all candidates’ expected utilities, then the objectives of the social planner and the can-

didates are well-aligned. However, they could differ in their risk preferences. The social planner seeks the optimal outcome and therefore tends to be risk-neutral. Candidates, on the other hand, could be risk-averse (e.g. when one’s health is poor), risk-neutral, or risk-seeking (e.g., if a candidate’s health is excellent, the person can take some risk of waiting for a long time to get a high-quality kidney).

Although the provider is interested in the best transplant decision for candidates, the decision to accept donor kidneys might be influenced by other factors as well. These factors may include benefits associated with transplantation, such as financial motivation, learning by experience, and innovation. Such factors may contain enough incentives to create a misalignment between the candidate’s objectives and the provider’s incentives [50].

While there could be some potential conflicts of interest, the relationship between candidates and the service provider is typically cooperative. We capture this unique and interesting relationship in our model with a spectrum of the service provider’s behavior: from self-interested (or rational) where the provider only takes her own objectives into account, to partially/completely altruistic where the provider aligns her decisions with the candidate. In the next subsection, we first model the candidate’s utility. Section 4.2 provides model details when providers are partially altruistic, making decisions to maximize a combination of patients’ and their own utility.

3.3 Utility function

We quantify the candidate’s health as the remaining lifetime on dialysis. We use $h_i > 0$ to denote the remaining lifetime on dialysis of candidate i when the person joins the active set, and it is evaluated by the doctor.

- Lemma 3.1** 1. Let $k_{c;i}$ be the acceptance threshold selected by candidate i . Given the assignment policy $\vec{p}$, the arrivals of acceptable kidneys to candidate i follow a Poisson process with a rate of $\mu_{a;i} = \mu_{o;i} F_Q(k_{c;i})$ where $\mu_{o;i} = p_i \mu$ is the offer rate of candidate i determined by the assignment policy $\vec{p}$.
2. Candidate i ’s waiting time from joining the active set A until the transplant, $w_{t;i}$, is the realization of an exponentially distributed random variable with rate $\mu_{a;i}$, i.e., $W_{t;i}(k_{c;i}) \sim \text{Exp}(\mu_{a;i})$

The probability that the candidate has to wait at least $w_{t;i}$ is given by $P(W_{t;i}(k_{c;i}) \geq w_{t;i}) = F_{W_{t;i}(k_{c;i})}(w_{t;i})$. We can find the $w_{t;i}$ which will not be exceeded with probability $\eta_{c;i}$. Then,

$$w_{t;i} = F_{W_{t;i}(k_{c;i})}^{-1}(\eta_{c;i}) = \frac{\ln \frac{1}{1-\eta_{c;i}}}{\mu_{o;i} F_Q(k_{c;i})} \quad (1)$$

⁷ Such assumptions are common practice in the literature for simplicity, even though they may not capture every real-world detail.

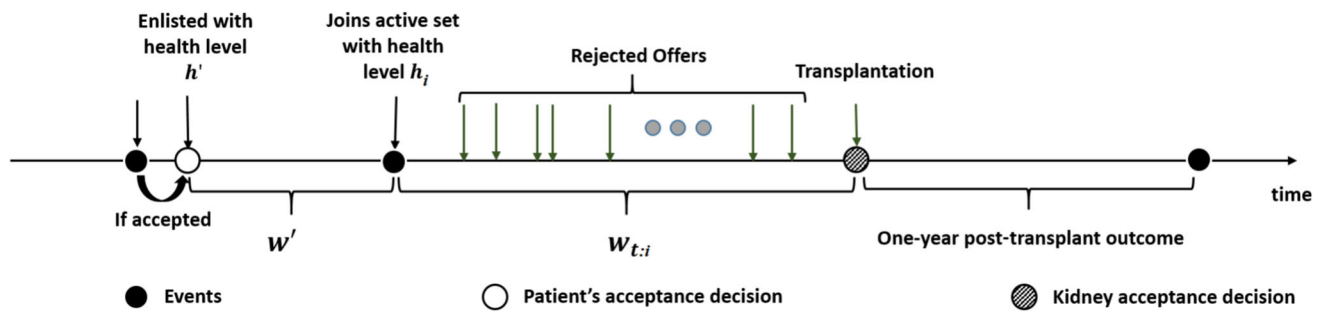


Fig. 2 Major stages of kidney assignment and transplantation process in our model. The transplant timeline starts with admitting the patient with a health level of h' to the waitlist by a transplant center. The patient has to wait for a while to join the active set and receive a donor kidney offer. After receiving an offer, the decision-makers, including doctors and the patient, evaluate the offered kidney. If the offered kidney's qual-

ity is below the decision maker's kidney quality threshold, the offer gets declined, and the patient has to wait for the next offer. However, if the offer is accepted, the transplant occurs, and the patient gets served. After the transplant, we consider one-year post-transplant outcomes to study the effect of performance measures and quality oversight policy

holds where $\eta_{c:i}$ captures the decision-maker's risk preference and increases as the decision-maker becomes more conservative and risk-averse in terms of wait time. For candidates, this probabilistic measure reflects the risk of deteriorating health; for centers, it means the risk of losing a patient while on the waitlist. Our analysis illustrates results for risk neutrality (i.e., $\eta_{c:i} = 1 - \frac{1}{e}$), meaning that $w_{t:i}$ is the expected time until a kidney of minimum quality $k_{c:i}$ is offered given by $E[W_{t:i}(k_{c:i})] = \frac{1}{\mu_{o:i} F_Q(k_{c:i})}$. The results can be easily extended based on different levels of risk preference among the decision-makers.

Candidate utility The candidate's utility includes the utility while being on dialysis (pre-transplant utility) and the post-transplant utility. For candidate i , time on dialysis includes the waiting time from joining the active set A until the transplant time $w_{t:i}$ and the time back on dialysis $w_{p:i}$ after the transplanted organ fails. Recall that we define h_i as the health of a candidate i as the expected remaining time on dialysis without a transplant. Since both times are realizations of random variables, we have the following for the total time on dialysis:⁸

$$h_i = E[W_{t:i}] + E[W_{p:i}] \quad (2)$$

Candidate i 's total utility U_i^t consists of her pre-transplant, post-transplant with working organ, and post-transplant after kidney-failure utilities. The candidate's total utility in

quality-adjusted life years is

$$U_i^t = \alpha U_i^{pre-transplant} + \beta U_i^c + \alpha U_i^{post-transplant failure} \quad (3)$$

where $U_i^{pre-transplant}$, U_i^c , and $U_i^{post-transplant failure}$ are the unadjusted utilities measured in years for pre-transplant, post-transplant with working organ, and post-transplant after kidney-failure utilities, respectively. The adjustment factors α and β represent the estimated quality of life scores without and with receiving kidney transplantation, respectively. Wit et al. [52] quantify these scores as $\alpha = 0.75$ and $\beta = 0.90$.

While the candidate i waits on dialysis in the pre-transplant and post-transplant kidney failure stages, she can receive a total unadjusted utility of D_i determined by her health condition h_i . If the candidate waits $w_{t:i}$ for a transplant, then $U_i^{pre-transplant} = w_{t:i}$ and $U_i^{post-transplant failure} = D_i - w_{t:i}$. The candidate realizes U_i^c if she receives a kidney transplant. We model the candidate's post-transplant utility, $U_i^c(\cdot)$ for a candidate with health h_i who receives a kidney of quality q as

$$U_i^c(h_i, q, w_{t:i}) = B(h_i, q) [1 - C(h_i, w_{t:i})] \quad (4)$$

where $B(\cdot)$ is the benefit the candidate receives from the transplant. The benefit increases with candidate health h_i and the kidney quality, i.e., $\frac{\partial B(h_i)}{\partial h_i} > 0$ and $\frac{\partial B(q)}{\partial q} > 0$. $C(\cdot) \in [0, 1]$ represents the candidate's health deterioration factor after a wait time of $w_{t:i}$ starting when the candidate joins the active set. The health deterioration strictly increases with the wait time and $C(\cdot)$ becomes 1 when the candidate is expected to die, i.e., when the wait time $w_{t:i}$ reaches the expected remaining lifetime on dialysis without a transplant h_i .

⁸ Based on the comparable survival outcomes observed in patients returning to dialysis after kidney transplant failure and those awaiting their first transplant [51], we assume that the wait times for the two groups are likely to be within a similar range. Hence, we approximate $W_{p:i}$ using $W_{t:i}$. While the approximation is based on prior evidence, we acknowledge that variability in patient experiences could lead to estimation errors. Our numerical estimation considers adult first-time kidney transplant recipients in the United States for this approximation.

The realized utility is uncertain, and we can define the total expected utility as:

$$E[U_i^t] = \alpha E[U_i^{pre-transplant}] + \beta E[U_i^c] + \alpha E[U_i^{post-transplant failure}] \quad (5)$$

where $E[U_i^{pre-transplant}] = E[W_{t:i}(k_{c:i})]$ and, $E[U_i^{post-transplant failure}] = D_i - E[W_{t:i}(k_{c:i})]$ are expected utilities that the candidate receives with respect to the expected wait time. The expected utility for selecting a kidney of quality $q \geq k_{c:i}$ is then

$$E[U_i^c] = \frac{\int_{k_{c:i}}^1 U_i^c(h_i, q, w_{c:i}) f(q) dq}{\int_{k_{c:i}}^1 f(q) dq} \quad (6)$$

To analyze the properties of the expected utility, we define functional forms for $B(h_i, q)$ and $C(h_i, w_{t:i})$ in U_i^c . Although any increasing function serves to demonstrate the idea that the benefit from transplantation increases as the kidney quality increases, we use a linear form here for the sake of model tractability. Let

$$B(h_i, q) = m_{h_i} q \quad (7)$$

where m_{h_i} indicates the best outcome of a candidate with health level h_i , which occurs when the highest possible quality kidney is transplanted to her immediately after joining the active set. To calculate the expected utility with respect to the expected wait time, we need to estimate the cost of deterioration $C(h_i, w_{t:i})$, which is a function of the candidate's health and the time the candidate has to wait for the transplant. This cost is given by

$$C(h_i, w_{t:i}) = \frac{1}{h_i} w_{t:i}. \quad (8)$$

If the wait time is zero, then the candidate's health deterioration is zero. The candidate's maximal health deterioration occurs when the wait time approaches h_i , i.e. the expected remaining lifetime on dialysis.

4 Kidney acceptance decisions

Several parties are involved in the kidney transplantation process. In this section, we derive optimal organ acceptance decisions (i.e., quality thresholds) for the two parties involved in making the acceptance decision, namely the candidate and the service provider, and compare them to the outcome desired by the social planner, who is interested in maximizing the utility created by the whole transplantation system.

4.1 Candidate's optimal decision

Each candidate tries to use a decision strategy to optimize her total expected utility, $E[U_i^t]$ (given in Eq. 5). Note that

$$\begin{aligned} & \alpha E[U_i^{pre-transplant}] + \alpha E[U_i^{post-transplant failure}] \\ &= \alpha E[W_{t:i}(k_{c:i})] + \alpha (D_i - E[W_{t:i}(k_{c:i})]) = \alpha D_i, \end{aligned} \quad (9)$$

which does not depend on the kidney quality q , and thus can be omitted from the optimization problem. Therefore, to find an optimal threshold strategy $k_{c:i}^*(h_i)$ for a candidate with health h_i , we only need to consider $E[U_i^c]$ in the optimization problem, which can be formulated as:

$$\begin{aligned} & \max_k E[U_i^c] \\ & \text{s.t. } 0 \leq k_{c:i} \leq 1 \end{aligned} \quad (10)$$

where $E[U_i^c]$ is the candidate's total expected utility to accept a kidney of minimum quality $k_{c:i}$.

Proposition 1 *The optimal candidate's kidney acceptance threshold, $k_{c:i}^*$, under different risk preferences of the candidate expressed by $\eta_{c:i}$ is*

$$k_{c:i}^* = \begin{cases} 1 & \text{if } k_{c:i}^* > 1 \\ 0 & \text{if } k_{c:i}^* < 0 \\ 1 - \sqrt{\frac{2 \ln\left(\frac{1}{1-\eta_{c:i}}\right)}{\mu_{o:i} h_i}} & \text{otherwise} \end{cases} \quad (11)$$

The optimal threshold in Eq. 11 is influenced by both donor kidney supply rate ($\mu_{o:i}$) and the candidate's health (h_i). A healthier candidate (higher h_i value) demands a higher quality kidney. Moreover, in case of an increase in kidney supply, which results in an increase in p_i , and ultimately a higher offer rate, the candidate would become more selective regarding the kidney quality. A risk-averse candidate (with $\eta_{c:i} < 1 - \frac{1}{e}$) has a tendency to set a lower $k_{c:i}^*$ threshold reducing the risk of long waiting time, while a risk-seeking candidate (with $\eta_{c:i} > 1 - \frac{1}{e}$) is willing to wait longer for a better kidney. Our findings regarding candidates' acceptance decisions and the impact of allocation policies on outcomes are consistent with findings of the prior literature on the topic [e.g., see 38, 53].

4.2 Service provider's optimal decision

Given the unique and sophisticated relationship between the service provider and the candidates enrolled in his transplantation program, we model the service provider in a range from completely self-interested to completely benevolent. While a candidate decides a quality threshold $k_{c:i}$ that optimizes her expected utility, the service provider's decision variable is a recommended quality threshold $k_{p:i}$ for candidate i .

Given the medical expertise, the provider's recommendation can influence the candidate's decision. In the following analysis, we solve the service provider's optimization problem under the assumption that the candidate follows the provider's recommendation.

Completely self-interested provider Let $U_i^p(\cdot)$ be the utility that a transplantation for candidate i generates for the service provider. B^p is a fixed benefit (e.g., monetary) for the transplant. A self-interested provider has the incentive to perform a transplant and thus receive this benefit before a candidate departs the system due to death on the waitlist or being delisted for other reasons. Therefore, the provider's expected utility when using the threshold $k_{p:i}$ for a candidate with health h_i is

$$E[U_i^p] = B^p P(W_{t:i}(k_{p:i}) \leq h_i) = B^p \int_0^{h_i} \mu_{o:i}(1 - k_{p:i}) \exp(\mu_{o:i}(1 - k_{p:i})t) dt, \quad (12)$$

where the random variable $W_{t:i}(k_{p:i})$ represents the expected wait time for a kidney with a quality of at least $k_{p:i}$. As shown in Lemma 3.1, the wait time $W_{t:i}(k_{p:i})$ has an exponential distribution with parameter $\mu_{o:i}P(Q \geq k_{p:i})$ and depends on the provider's threshold $k_{p:i}$. The provider's optimal threshold is denoted by $k_{p:i}^*$. The service provider's total expected utility is

$$U_i^p(\vec{k}_p|\vec{p}) = \sum_{i=1}^N p_i E[U_i^p] = B^p \sum_{i=1}^N p_i P(W_{t:i}(k_{p:i}) \leq h_i) \quad (13)$$

Given the assignment policy $\vec{p}$, $U_i^p(\vec{k}_p|\vec{p})$ is maximized when $E[U_i^p]$ is maximized $\forall i$. We can show that:

Proposition 2 *A self-interested provider maximizes his total expected utility by setting the kidney quality threshold to the lowest possible kidney quality, i.e., by accepting any kidney.*

Clearly, the extreme case of completely self-interested providers as described above does not exist. Providers care about their patients and their health outcomes. Next, we will discuss a more realistic provider.

Self-interested vs. altruistic providers At the other extreme, a completely altruistic provider would strictly choose the quality threshold that optimizes the patient's utility. A realistic situation is a provider that acts somewhere between both extremes. Such a not-completely self-interested (general) provider is balancing the two competing objectives of completely self-interested and altruistic providers. To reflect this fact, the general provider's expected post-transplant utility can be defined as a weighted average of these two objective

functions.

$$E[U_i^{gp}] = \tau E[U_i^c] + (1 - \tau) E[U_i^p], \quad (14)$$

Therefore the decision threshold $k_{t:i}$ applied by a general provider falls in the interval of $[k_{p:i}^*, k_{c:i}^*]$ or in the other words:

$$k_{t:i} = \delta k_{c:i}^* + (1 - \delta) k_{p:i}^*, \quad (15)$$

where δ represents the provider's degree of benevolence in the range $0 \leq \delta \leq 1$.

Here, the general provider can choose, depending on his incentive, to obtain maximum benefit from transplant (i.e., completely self-interested with $\delta = 0$) as opposed to contributing to the best social outcome (i.e., completely altruistic with $\delta = 1$). The general provider chooses $k_{t:i}$ depending on how much utility he is willing to give up ($E[U_i^p(h_i, k_{p:i}^*)] - E[U_i^p(h_i, k_{t:i})]$) in order for the candidate to gain utility ($E[U_i^c(h_i, k_{t:i})] - E[U_i^c(h_i, k_{p:i}^*)]$). Consistently, [53] show surgeons' trade-off when making organ acceptance decisions where they turn down low-quality organs in the hope of seeing better organ offers and therefore a higher utility for the patient.

4.3 Social welfare and social planner

The social planner is interested in maximizing social welfare, which means optimizing the whole system in terms of realized utility. The system contains the candidates, the providers, and the social planners (payers, see the supplementary document on kidney transplant background). Since the service provider's financial benefits are just a transfer within the system (from payers to providers), we can concentrate on the candidates' utility.

Lemma 4.1 *The social welfare utility function is maximized if and only if each candidate's individual utility is optimized under risk neutrality. The socially optimal kidney quality threshold, $k_{s:i}^*$, that maximizes the expected utility of a candidate with health h_i under risk-neutrality is*

$$k_{s:i}^* = 1 - \sqrt{\frac{2}{\mu_{o:i}h_i}} \quad (16)$$

If the candidate is risk-neutral and the provider is completely altruistic, then all involved parties have the same optimal decision thresholds $k_{s:i}^* = k_{c:i}^* = k_{t:i}$. This indicates that the incentives are completely aligned, and no additional intervention like quality oversight would be needed. However, if the provider is not entirely altruistic and accepts organs that are of lower quality with $k_{t:i} < k_{s:i}^*$, then the

candidate and social planner lose utility. Given the fact that there are complex trade-offs and expert opinion is critical in accepting a kidney offer, it is reasonable to assume that many candidates will trust the recommendation of their providers [54].

When the candidate entrusts the decision to the provider and the provider is not entirely altruistic, social welfare is not maximized resulting in a social loss. This loss happens because incentives are not aligned between the provider and the candidate or social planner. In order to re-align incentives and reduce social loss, the social planner can use performance assessment and related policy tools we will discuss next.

5 Optimal quality oversight

In this section, we identify factors affecting quality oversight and investigate the relationship among those factors to introduce an optimal citation policy (CP). We demonstrate how a CP influences the kidney acceptance decision of providers, i.e., how it changes $k_{p:i}^*$. An optimal citation program must be strong enough to encourage the provider to prioritize candidates' outcomes from transplantation over all other potential incentives. That is, it needs to create an incentive structure that moves $k_{p:i}^*$ as close as possible to $k_{c:i}^*$. Then the patient and the social planner do not need to rely on the degree of benevolence by the provider. However, a poorly calibrated, too-strict CP can also create risk-averse behavior, which might lead providers to dissuade candidates from accepting beneficial organs (i.e., produce a $k_{p:i}^*$ that is much larger than $k_{c:i}^*$) or to limited access to care for sicker ESRD candidates.

Since the existing CPs are typically based on the post-transplant survival of the candidate or the kidney, we first introduce a post-transplant survival model.

5.1 Post-transplant survival

We define a continuous random variable T to represent the time until the occurrence of an event of interest to the regulator, such as post-transplant kidney failure or candidate loss. We collectively refer to such events as failures or transplant failures for brevity. We define $g(t)$ and $G(t)$ to be the probability density function and the cumulative distribution function of T , respectively. The standard survival function is defined as

$$S(t) = P(T > t) = 1 - G(t) = \int_t^\infty g(x) dx, \quad (17)$$

where $S(t)$ is the probability of no failure up to time t . The hazard function or instantaneous rate of occurrence of the

event is

$$\lambda(t) = \lim_{dt \rightarrow 0} \frac{P(T \leq t + dt | T \geq t)}{dt} \quad (18)$$

$S(t)$ written in terms of $\lambda(t)$ is $S(t) = \exp(-\lambda(t))$. The numerator in Eq. 18 is the conditional probability that the failure occurs in the time interval $[t, t + dt)$ given that it has not happened before. Using a constant risk over the first one-year post-transplant period for a kidney of quality $q \geq k_i$ (we use k_i to reflect that it could be the threshold of any player), we define the hazard function as $\lambda_{q \leq k_i}(k_i, h_i, t) = \lambda_{q \leq k_i}(k_i, h_i)$ for all $t \leq 1$. The reason that t is dropped from the hazard function is that we assume constant risk over a relatively short period of one year.

The hazard function, $\lambda_{q \leq k_i}(k_i, h_i)$ decreases in both patient health h_i and kidney acceptance threshold k_i , which reflects the fact that as the patient's health and the quality of the accepted kidney decrease, the risk of post-transplant failure increases. We introduce a multiplicative functional form for $\lambda_{q \leq k_i}(k_i, h_i)$ as $\lambda_{q \leq k_i}(k_i, h_i) = \lambda'(h_i)(1 - k_i)$, where $\lambda'(h_i)$ is a hazard factor that decreases in patient health. The resulting survival function for one-year post-transplant survival is

$$S_i(k_i, h_i, t) = \exp(-\lambda'(h_i)(1 - k_i)) \quad \text{for all } t \leq 1, \quad (19)$$

and the failure function (i.e., the probability of failure) is defined as $F_i(k_i, h_i, t) = 1 - S_i(k_i, h_i, t)$.

5.2 Provider's decision policy under quality oversight

For a given organ assignment policy $\vec{p} = [p_i]$, we model how a post-transplant survival-based citation strategy imposed by a regulator impacts the provider's optimal acceptance threshold for a candidate $k_{p:i}$. The total expected net provider utility of performing a transplant under a given citation policy CP is

$$E[U_i^P] = [B^P - C_p P(\text{citation})] P(W_{t:i}(k_{p:i}) \leq h_i), \quad (20)$$

where B^P is a fixed benefit (net of cost) for the transplantation, C_p is the cost of being cited as low-performing by the regulator, and $P(\text{citation})$ is the estimated probability of citation as a result of performing the transplant. $P(W_{t:i}(k_{p:i}) \leq h_i)$ is the probability that the candidate receives a timely kidney offer of acceptable quality under the threshold choice of $k_{p:i}$.

Next, we need to find the probability of being cited. The regulator will only cite a provider if an adverse event, here called a failure, occurs. However, not all failures will lead

to being cited (e.g., the loss of a single very sick patient may not result in a citation). The probability of being cited can be expressed by Bayes' theorem as $P(\text{citation}) = P(\text{citation} | \text{failure}) P(\text{failure}) / P(\text{failure} | \text{citation})$. Since the provider is only cited for the candidate if a failure occurs, the following always holds: $P(\text{failure} | \text{citation}) = 1$. A failure occurs when the post-transplant survival is less than a year, leading to the failure probability defined as $P(\text{failure}) = F_i(k_{p:i}, h_i, t)$. We denote $f_p = P(\text{citation} | \text{failure})$ as the probability of being cited given a failure.

Assumption 5.1 The provider will not set a kidney quality threshold $k_{p:i}$ that makes the expected net benefit of the transplantation negative, i.e., the transplantation would reduce the provider's expected overall profit.

Proposition 3 Under Assumption 5.1, the provider's optimal kidney quality threshold under quality oversight ($k_{p:i|CP}^*$) is

$$k_{p:i|CP}^* = 1 + \frac{1}{\lambda'(h_i)} \ln \left(1 - \frac{B^p}{C_p f_p} \right) \quad (21)$$

The regulator's oversight policy can utilize one or both of the following two tools: (i) the cost impact of CP represented by C_p , and (ii) the likelihood that a failure will be considered a citation event f_p . Thus, the regulator can influence the optimal threshold $k_{p:i|CP}^*$ chosen by the provider. According to Eq. 21, C_p and f_p have the collective multiplicative effect of $C_p f_p$, and an increase will increase the provider's optimal threshold. Next, we will define the optimal CP and discuss the role of C_p and f_p .

5.3 Optimal citation mechanism

The optimal citation mechanism needs to be designed to incentivize the provider to choose for each candidate the socially optimal kidney quality threshold, that is $k_{p:i|CP}^* = k_{s:i}^*$.

Proposition 4 The optimal value of $C_p f_p$ denoted as $[C_p f_p]^*$ to achieve $k_{p:i|CP}^* = k_{s:i}^*$ for a single candidate is

$$[C_p f_p]^* = \frac{B^p}{1 - \exp \left(-\lambda'(h_i) \sqrt{\frac{2}{\mu_{o:i} h_i}} \right)} \quad (22)$$

The cost of citation that the regulator imposes on a provider, C_p , includes the cost for the personnel and the time needed to go through a quality review or a systems improvement agreement. The cost can also include the lost future revenue in case a provider is excluded from performing organ transplants. Alternatively, C_p could also be implemented as a direct fine or repayment of received transplantation fees. The severity and amount imposed by C_p are under the regulator's control. The probability of getting cited in case of

an observed failure, f_p , represents how strict the CP is. In the most extreme case of $f_p = 1$, every adverse event will lead to a citation and C_p will be imposed. The optimal value of the expected cost impact of citation policy, $[C_p f_p]^*$, represents all possible combinations of C_p and f_p for which $C_p f_p = [C_p f_p]^*$, i.e., the same impact can be achieved with every combination of C_p and f_p that keeps the product constant. This offers the regulator a range of equivalent CPs, from citing more frequently at a lower cost per citation to citing less frequently at a higher cost per citation. Equation 22 also shows that as the provider's benefit B^p per transplant increases, the regulator needs to adjust its strategy accordingly to avoid deviation of the provider's optimal decision from the social optimum. Furthermore, according to Eq. 22, increasing h_i requires a higher value for $[C_p f_p]^*$. This trade-off corresponds to the need for risk adjustment, i.e., the optimal CP needs to penalize providers more for the loss of healthier candidates than the loss of sicker ones. The insight is intuitive because the likelihood of post-transplant failure (e.g., mortality rate) among the high-risk population is higher than that of the low-risk population. Similarly, in Eq. 22, increasing $\mu_{o:i}$ increases $[C_p f_p]^*$. As the kidney supply increases, the regulator can expect to see better outcomes since more choices would be available to the candidate. Also, there would be a wider range of options from which the transplantation center can select desired kidneys. Next, we define and examine the cases of insufficiently intense and over-intense CPs.

Corollary 5.1 If a given CP is insufficiently intense, i.e., $C_p f_p \leq [C_p f_p]^*$, altruistic providers will keep their kidney quality threshold at $k_{s:i}^*$, and non-completely altruistic providers will set their optimal kidney quality thresholds in the range of $[k_{p:i}, k_{s:i}^*)$. If a given CP is over-intense, i.e., $C_p f_p > [C_p f_p]^*$, all providers will set their kidney quality threshold in the range of $(k_{s:i}^*, 1]$.

The result above suggests that while insufficiently intense CPs affect only providers that are not completely altruistic, over-intense CPs affect all providers. The finding supports the argument that overly strict report cards will have a negative effect on transplantations. Suppose the citation policy does not take supply and demand in the region of the provider into account. In that case, an over-intense citation policy is more likely to be observed among transplant centers that have a high demand-to-supply ratio. A lower deceased donor organ donation rate (supply) is one of the essential elements that influence the demand-to-supply ratio. The second element is the number of waitlisted candidates (demand). Based on Proposition 4, we see that given a fixed citation cost C_p , transplant centers that are located in regions with lower individual offer rates ($\mu_{o:i}$) are at risk of being cited more frequently (increased f_p). To reduce their risk of being cited, we should

see the adverse effect of the CP that these centers will perform fewer transplantations and only use higher-quality kidneys.

6 Impact of changes to the kidney allocation system

In prior sections, we have identified how supply (the offer rate μ) and demand (reflected in the length of the waiting list) impact the relationship between the quality oversight via CP and the providers' choices for kidney acceptance. Supply for individual candidates is affected by the kidney allocation system, KAS, represented by the allocation probability vector $\vec{p}$ since the individual offer rate for candidate i is $\mu_{o:i} = p_i \mu$. Since this rate affects the optimal decision, it also influences the optimal CP.

Motivated by the changes recently introduced in the U.S. by OPTN to the kidney allocation system, we analyze the impact of such changes on the optimal kidney transplant decision. We also investigate changes the regulatory agencies need to make to adapt their optimal citation policy as a result of the new allocation rules. In the rest of this paper, we refer to the new kidney allocation system as post-KAS or the new KAS and refer to the old allocation system as pre-KAS or the old KAS.

6.1 Kidney transplant decision

Under the old KAS, all candidates had equal access to all kidneys. The new KAS implements an access policy with preferential treatment of certain patient groups. Under the new KAS, the top 20% of kidneys with kidney-donor profile index KDPI $\in [0, 0.2]$ are offered first to the top 20% healthiest candidates with the estimated post-transplant survival EPTS $\in [0, 0.2]$. From the social planner's perspective, the policy entails using the best quality kidneys for the healthiest candidates with the expectation that it will result in better outcomes in terms of the overall utility for the whole system.

To model this system, we classify kidneys and candidates into low- and high-quality classes by using the introduced KAS threshold of $\pi = 0.8$. In our model, the top-quality kidneys fall in the range of $q \in [\pi, 1]$. The kidney quality q measured as $1 - \text{KDPI}$ has a close to uniform distribution following the construction of KDPI. We assign all recovered kidneys with quality $q \geq \pi$ (i.e., top 20% of kidneys or low-risk kidneys) to the low-risk class. We also identify the $1 - \pi$ healthiest candidates to separate low-risk and high-risk candidates. We modify our model to accommodate the new allocation policy to show the impact of exclusively assigning the top 20% of kidneys (low-risk kidneys) to the top 20% of candidates (low-risk candidates).

6.1.1 Low-risk candidates

The kidney arrival process is modeled as a Poisson process with an arrival rate of μ . Top-quality (low-risk) kidneys arrive with a rate of $\mu_l = \mu P(Q \geq \pi)$ and low-quality (high-risk) kidneys arrive with a rate of $\mu_h = \mu P(Q < \pi)$. Under the new KAS, low-risk kidneys are exclusively offered to low-risk candidates and are shared among the $N_l = (1 - \pi)N$ low-risk candidates in the active set. The low-quality kidneys get assigned to either one of the N_l active low-risk candidates or one of the $N_h = \pi N$ active high-risk candidates.

To model the kidney acceptance decision by the low-risk candidate, we assume that low-risk kidneys have such high quality that any low-risk candidate and center will accept them for transplantation (we confirm this assumption is reasonable for a physician with expertise in transplantation). Therefore, low-risk candidates in the post-KAS period choose a threshold $k_l \leq \pi$. In order for high-risk candidates to have a chance to receive an organ, they will not demand a better kidney than low-risk candidates resulting in $k_h \leq k_l$ where k_h is the high-risk candidate kidney quality threshold post-KAS. The optimization constraint representing the second threshold is always met by our previous model, and we will show that it is also automatically satisfied by the new KAS model. The rate with which acceptable kidneys ($q \geq k_l$) are offered to low-risk candidates can be calculated as

$$\mu_l^a = \frac{\mu_l}{N_l} P(Q_{\geq \pi} \geq k_l) + \frac{\mu_h}{N_l + N_h} P(Q_{< \pi} \geq k_l) \quad (23)$$

Proposition 5 *The optimal kidney quality threshold of a low-risk candidate under the new KAS is*

$$k_l^* = 1 + \pi - \sqrt{\frac{1 + 2\pi}{\frac{\mu}{N} h_i}} \quad (24)$$

Based on this threshold post-KAS, the following corollary holds.

Corollary 6.1 *Under the new KAS implementation, the low-risk candidate increases the kidney quality threshold k_l^* as compared to the pre-KAS optimal kidney quality threshold k_s^* .*

The justification for this behavioral change is given by the fact that the new KAS limits the competition for top-quality kidneys to only low-risk candidates while these candidates still have in addition access to low-quality kidneys. In the new setting, the increase in the offer rate for low-risk candidates results in an effective increase in the supply for her. Thus, a candidate advantaged by the system will demand a higher kidney quality.

6.1.2 High-risk candidates

Under the new KAS, higher-risk candidates effectively lose access to high-quality donor kidneys. The arrival rate of the remaining kidney type is $\mu_h = \mu P(Q < \pi)$. Since all N active candidates get offers for these kidneys, the offer rate is $\frac{\mu_h}{N}$. The acceptance rate can be defined as

$$\mu_h^a = \frac{\mu_h}{N} P(Q_{<\pi} \geq k_h) = \frac{\mu\pi}{N} (\pi - k_h) \quad (25)$$

Proposition 6 *The optimal kidney quality threshold of a high-risk candidate under KAS is*

$$k_h^* = \pi - \sqrt{\frac{2}{\frac{\mu}{N}h_i}} \quad (26)$$

High-risk candidates respond to a threshold-based KAS by also accepting kidneys of lower quality than before because top-quality kidneys are no longer accessible to this group. The report published in [55] shows that after the introduction of a threshold-based KAS with $\pi = 0.8$, the kidney discard rate decreased. The model comes to the same conclusion since higher-risk candidates reduce their kidney quality requirements and will now accept kidneys that would have been discarded before.

6.2 Citation policy

The regulator needs to take into consideration that after implementing KAS, the assessment criteria for evaluating transplant program performance needs to be adjusted. As shown in Section 6.1, the optimal transplant-related decisions have changed due to the effect of the introduction of the new KAS on kidney supply. These decisions have a major impact on post-transplant results. In our citation model, the total expected cost of impact (i.e., the cost and the probability of citation given failure) depends on the kidney supply, individual offer rate, and social optimum. Since all these three factors have changed with the new KAS, the optimal combination $[C_p f_p]^*$ also needs to be appropriately adjusted to align providers' incentives with candidates' optimal outcomes under KAS.

Proposition 7 *The optimal citation policy under the new KAS depends on the candidate's risk group. The optimal citation policies for low- and high-risk candidates are as follows:*

$$[C_p f_p]^{*lkas} = \frac{B^p}{1 - \exp\left(-\lambda'(h_i)\left(\sqrt{\frac{2\pi+1}{\mu_o h_i}} - \pi\right)\right)} \quad (27)$$

$$[C_p f_p]^{*hkas} = \frac{B^p}{1 - \exp\left(\frac{-\lambda'(h_i)}{\pi} \sqrt{\frac{2}{\mu_o h_i}}\right)} \quad (28)$$

As a result, the risk-adjustment outcome should be modified in such a way that the provider is penalized by either lower cost or a lower chance for making an unsuccessful graft to a high-risk candidate. In other words, the CP should follow the underlying changes in KAS (allocation based on the risk profile of the candidates) to produce socially optimal outcomes. At the time of writing this manuscript, the CP does not account for the changes in KAS.

7 Candidate admission policy

Another important step that influences the transplantation process is the admission policies that providers follow in accepting patients to their centers. To be eligible for transplantation, the candidate needs to be referred to a transplant center by a nephrologist. The transplant team evaluates the candidate's overall health. If the patient is identified to be eligible for transplantation, she will be enlisted in both the transplant center's and the national waitlist. In addition to meeting the eligibility criteria for being waitlisted, the provider's expectations about the wait time and transplantation outcomes can influence the candidate acceptance decision [56, 57]. We model a rational service provider who will accept a patient if the expected benefit of having the patient outweighs the expected cost.

When a patient is referred to the transplant center (service provider), the state of the system is l (i.e., there are l patients in the system). The service provider evaluates the patient's health. We measure the patient's health as h' , which represents the time the patient can live on dialysis without a transplant (equivalently, the maximal survival time on the wait list). The service provider is not concerned with the lack of patients in their admission control policy. The reason is that the demand for kidney transplants is several times larger than the kidney supply, and there is consistently a long list of patients waiting for kidney transplants. In this situation, the service provider needs to weigh the chance that the patient survives until transplantation against the risk of losing the patient while on the waiting list due to death or disqualification. If the patient is admitted, there will be two time frames: (i) the waiting time until the patient makes it to the active set and starts receiving offers, and (ii) the waiting time from entering the active set until an organ is accepted.

If the patient is admitted when there are l patients ahead, the patient will move up the queue to the position $l - 1$ with rate $\mu(l)$, which is the sum of the leaving rate of the l patients ahead in the queue and the transplantation rate. A model to capture the complete dynamics of the system is obviously neither tractable nor practical since the leaving rate of the l patients depends on the health vector of these patients and the transplantation rate depends on the allocation policy and the health vector of the current population of the active set.

This would lead to a complicated continuous-time dynamic model. However, the average rate at which patients leave the queue depends on the queue length. To keep the model tractable, we model the queue dynamics at an aggregate level that can be estimated from the data as in the following.

Given the current state of the queue is l , the patients in the queue leave the system (due to deaths on the waiting list, disqualification, etc.) according to a Poisson process with rate lp_e where p_e is the probability that an arbitrary patient leaves the queue. The probability p_e can be estimated as the average rate of patients leaving during an observation period. For brevity of formulation and analysis, we consider a self-interested service provider where the threshold of acceptable kidney quality is $\underline{k}$, and the transplantation rate is $\mu \overline{F_Q}(\underline{k})$. If the patient is admitted when there are l patients in the system, the moving up rate to the position $[l - 1]$ could then be calculated as $\mu(l) = lp_e + \mu \overline{F_Q}(\underline{k})$. A service provider can use past data to estimate the long-term transplantation rate consistent with this formulation.

Let $W'(l, \mu)$ be the time duration from admission to entering the active set when the queue size is l and the kidney arrival rate is μ . Also, let $T_m(\mu)$ be the time it takes for the patient to move from position m to position $(m - 1)$, which has an exponential distribution with rate $\mu(m)$. Defining $l_c = (l - N) \vee 0$, we have:

$$W'(l) = \sum_{l_c}^l T_m(\mu). \quad (29)$$

If the patient makes it to the active set, the person's health level will then be $H = (h' - W'(l)) \vee 0$.

The waiting time from entering the active set until transplantation is W_t . As analyzed before, this duration depends on the patient health when entering the active set and the health levels of other patients in the active set, kidney supply rate μ , and the allocation policy. For admission to the waitlist, assume a uniform allocation policy. Under such a policy, the distribution of W_t is exponential with the rate of $\frac{1}{N} \mu \overline{F_Q}(\underline{k})$. For clarity in this section, we breach the notation and use $W_t(\mu)$ to indicate the dependence on the kidney supply rate. Let $S(h', l, \mu)$ be the survival probability until transplantation of a patient admitted with health h' when the system state is l . Then, we have:

$$S(h', l, \mu) = Pr(W'(l, \mu) + W_t(\mu) \leq h'). \quad (30)$$

The failure probability is: $1 - S(h', l, \mu)$. The optimal candidate's waitlisting decision for the center is to accept a candidate if the expected benefit of waitlisting her becomes greater than the expected cost of losing her before transplantation. Namely, the expected utility of the service provider

regarding the patient is not less than zero such that

$$E[U^P] = B'S(h', l, \mu) - C'(1 - S(h', l, \mu)) \geq 0, \quad (31)$$

where C' is the cost of waitlist mortality (e.g., reputation cost, losing public trust), and B' is the composition of two benefits, B and B^P . B is the benefit that the provider collects while taking care of waitlisted patients, and B^P is the payment made by CMS or the insurance company for transplantation.

Proposition 8 1. When there are l patients in the system, it is optimal for the service provider to admit a patient with the health score h' if $h' \geq h'_a(l, \mu)$, where $h'_a(l, \mu)$ solves the equation: $S(h'_a(l, \mu), l, \mu) = \frac{C'}{B' + C'}$
2. $h'_a(l, \mu)$ is increasing in l and decreasing μ

Proposition 8 characterizes the transplant center's optimal admission policy, a threshold policy based on patient health. The proposition suggests that the service provider assumes a more strict admission policy (a higher health threshold) if the length of the waitlist is higher and a more flexible policy (a lower threshold) if the kidney supply rate increases.

8 Empirical and computational analysis

We utilize data from UNOS, USRDS, and SRTR for model parameter estimation and numerical studies. The SRTR data contains detailed medical and demographic information for all donors, waitlisted candidates, and transplant recipients in the U.S., submitted by the members of OPTN between October 1987 and the end of 2022. The numerical illustration and data analysis conducted in this chapter are based on more than 400,000 patients who are first-time recipients of deceased donor kidney transplants. Our numerical results provide a comparison of kidney transplant outcomes before and after CoP implementation. More precisely, for providers, we illustrate kidney transplant outcomes by reporting an observed kidney acceptance strategy obtained from the data and computing the corresponding expected post-transplant utility. We then use our proposed model to compare the providers' outcomes with that of the socially optimum kidney acceptance strategy before and after CoP implementation. We also compute and compare the overall social loss, deficiencies, and degree of benevolence before and after CoP introduction. We present the percentage of patients who received suboptimal kidneys due to over-intense and insufficiently intense citation policies. Finally, we discuss the total life-years saved as a result of CoP implementation.

8.1 Effect of quality oversight on provider's transplant decision

On March 30, 2007, CMS issued regulations and CoP for hospital-based kidney transplant programs; the regulation took effect on June 28, 2007. The new requirements move Medicare-covered transplant programs toward an outcome-focused system that reflects the clinical experience, resources, and commitment of each transplant program. According to [58], between 2005 and 2014, the number of active candidates grew on average by 2.91% annually, which led to increases in the demand-to-supply ratio (DSR). As a result, the average observed wait time till kidney transplantation after 2007 over the entire population (11 regions) has increased by 123 days (from 587 to 710); this resulted in a reduction of waitlisted candidates' health equivalent to six months on dialysis on average.

Table 1 summarizes transplant outcomes (averages) before and after CoP implementation in 2007 for each region and compares the observed changes with the optimal social outcomes when the optimal threshold policies k_s^* are used. As a result of the candidate's health reduction due to longer wait time and higher DSR, the average social optimum kidney quality threshold given by the model after CoP introduction has decreased by 0.03 from 0.69. The reason for this reduction is based on the fact that our model considers the patient's health and DSR to determine the optimal kidney quality threshold.

One of the advantages of our model is that it provides a framework to evaluate the efficiency of CoP. Considering the dependency of the social optimum upon the availability of donor organs and total demand can help explain the observed utility reduction. Without taking into account the effect of kidney supply and demand on transplant decisions, it might

be difficult to capture the true effect of the transplant center's performance assessment in a dynamic environment where the waitlist may grow over time.

Building up on our model, we compare performances with respect to different metrics before and after CoP. Our approach to modeling patient and waitlist dynamics allows us to extract an appropriate and practical conclusion as we capture the effect of waitlist length on time to transplant (or wait times) and patient health. To assess the effect of quality oversight on the provider's response to kidney transplantation decisions and outcomes, we measure social loss, deficiency, and degree of benevolence for each candidate as follows. (i) The social loss attributable to a candidate is defined as the difference in the value of the social welfare utility function between the socially optimal kidney transplant decision and the provider's decision as observed in the data. (ii) Efficiency is measured as the expected utility under the provider's decision relative to the expected utility under the socially optimal decision, while deficiency is defined as one minus efficiency. (iii) The observed degree of benevolence reflects how closely the provider's decision aligns with the socially optimal decision. This is calculated by expressing the provider's threshold as a weighted combination of the purely self-interested provider threshold and the socially optimal threshold. For a purely self-interested provider, the threshold is zero, leading to a degree of benevolence calculated as the ratio of the provider's threshold to the socially optimal threshold. We provide the corresponding mathematical formulation in Appendix D.3.

Table 2 contains the average results of performance measurements before and after CoP implementation. Our analysis of providers' performance indicates that the introduction of the CoP has, on average, reduced the social loss and deficiency by almost six months and 31%, respectively, and

Table 1 Average transplant outcome statistics before and after the introduction of CoP (2007)

Region	Outcomes Before CoP				Outcomes After CoP			
	Observed	Utility	Social optimal	Utility	Observed	Utility	Social optimal	Utility
	k_t		k_s^*		k_t		k_s^*	
1	0.56	11.34	0.64	12.27	0.56	10.98	0.63	11.53
2	0.47	11.21	0.66	12.63	0.54	10.81	0.65	11.86
3	0.57	13.02	0.73	14.21	0.58	12.16	0.70	12.91
4	0.57	12.70	0.72	13.78	0.63	11.16	0.69	12.26
5	0.55	11.69	0.66	12.72	0.57	12.02	0.64	12.12
6	0.55	13.02	0.73	14.41	0.67	12.26	0.68	12.29
7	0.52	11.77	0.67	12.99	0.55	10.96	0.64	12.21
8	0.61	13.01	0.73	13.98	0.63	12.51	0.70	12.79
9	0.48	11.00	0.60	12.17	0.49	10.05	0.57	10.93
10	0.57	12.83	0.73	14.06	0.60	11.87	0.68	12.57
11	0.56	12.47	0.69	13.53	0.58	12.19	0.66	12.48
Average	0.55	12.19	0.69	13.34	0.58	11.54	0.66	12.18

Table 2 Provider's performance comparison

	Provider's Preference	Risk	Social Loss	Deficiency	Degree of Benevolence
Averages Before CoP Introduction	0.73		1.15	0.09	0.80
Averages After CoP Introduction	0.67		0.66	0.06	0.89
Changes	-8%		-43%	-31%	+12%

the degree of benevolence has increased by 12%. Also, we observe 43% reduction in social loss, and the provider's risk preference moved by 8% closer toward risk neutrality. This shows that the introduction of the CoP had a positive effect that was masked by the increase in the number of candidates waiting for transplants.

8.2 The optimal citation policy

In this section, we analyze how the current citation policy affects kidney acceptance decisions and assess its alignment with the optimal policy. The goal is to determine whether the current citation policy is close to the optimal and, if not, to measure the gap and identify how much adjustment is needed to achieve optimal performance (Table 3).

To quantify this, we define Δ , which represents the difference between the provider's expected cost under the current citation policy and the expected cost under the optimal policy. A Δ value close to zero indicates that the provider's perception of the citation policy aligns with the optimal, leading to threshold decisions consistent with the social optimum. We calculate Δ across 11 U.S. regions, highlighting regional differences in current and optimal citation policies following the implementation of the CoP. We provide the details of the estimation procedure in Appendix D.3.

Table 4 represents the average Δ of kidney transplant recipients over 11 regions. An over-intense CP leads to positive values of Δ , and an insufficiently intense CP is associated with negative values of Δ . We consider for this analysis the CP as sufficiently intense if Δ falls in the interval of $(-\$1000, \$1000)$. When $\Delta > \$1000$ for a provider, then the provider's response to the quality oversight results in choosing a kidney quality threshold above the socially optimal one, which leads to a sub-optimal outcome. In this case, the social planner needs to reduce the expected cost of unsatisfactory outcomes by Δ to avoid the negative effect of the provider's risk-averse decision. On the other hand, when $\Delta < -\$1000$ for a provider, the provider's decision is below the social optimum. To address such provider behavior, the social planner's imposed expected cost impact of CP has to be increased by Δ . Such an increase can be achieved by adjusting the frequency of citation which can factor in the availability of organs in each region.

Figure 3 shows, based on our analysis, the effect of CP due to CoP in each region. Given the recipients' health profiles

and kidney supply-demand in each region, on average, the current CoP has led to sub-optimal kidney acceptance decisions for 15% and 41% of recipients due to the over-intense and insufficiently intense oversight policy. In contrast, the optimal decisions have been reached only for the remaining 44% recipients. In addition, we conduct a descriptive analysis of organ discard rates and find evidence of a positive correlation between the intensity of oversight policies and organ discard rates (see Appendix D.4 for more details). Future research can further explore the precise relationship between CoP, organ supply, and transplant center behavior, and the resulting discard rates.

9 Concluding remarks

In this study, we have developed a model that can be used to analyze the design of optimal quality oversight mechanisms based on report cards. The model can be used together with observed data to clarify the effects of changes to the mechanism on provider behavior and subsequent outcomes. The primary motivation for studying such effects is the criticisms regarding the measuring and assessment of performance in healthcare, in general, and reported shortcomings and unintended consequences of report cards for kidney transplantation, in particular.

Specifically, we have modeled the care processes starting with the transplant centers' candidate selection all the way to the kidney acceptance decisions. Our model accounts for both the supply and demand factors for kidneys as well as the impact of quality oversight on provider decisions. Our structural analysis reveals how waitlist dynamics (e.g., those based on supply factors including region and blood type) and the underlying kidney allocation rules are critical in establishing a socially optimal quality oversight mechanism. Despite the difficulties in performing a direct empirical comparison of transplant decisions before and after the introduction of CMS's CoP in 2007, our modeling framework provides a tool to perform this comparison. Our modeling framework and the empirical analysis also shed light on how changes in the allocation system affect the acceptance of patients by transplant centers.

The novelty of our work is in its aggregation of determinants of quality outcomes in a unique healthcare context: processes of care leading to decisions (e.g., supply

Table 3 Optimal citation policy

Region	From Data		Calculated for Provider			Calculated for Social Optimal			Citation Policy					
	Offer Rate	h	Wait Time	k _t	Utility	PF	C _p f _p	Expected Cost	k _s [*]	Utility	PF	[C _p f _p] [*]	Set Expected Cost	Δ
1	6.00	5.80	733	0.56	10.98	0.051	\$262,600	\$6,319	0.63	11.53	0.047	\$291,922	\$5,500	-\$819
2	6.50	5.80	723	0.54	10.81	0.051	\$285,701	\$7,618	0.65	11.68	0.044	\$290,937	\$5,500	-\$2,118
3	8.70	5.69	687	0.58	12.16	0.052	\$274,061	\$8,250	0.70	12.91	0.032	\$370,546	\$5,500	-\$2,750
4	8.00	5.50	649	0.63	12.36	0.042	\$280,020	\$6,998	0.64	12.26	0.041	\$346,024	\$5,500	-\$1,884
5	5.00	5.85	857	0.57	11.16	0.037	\$267,276	\$6,988	0.64	12.12	0.041	\$318,205	\$5,500	-\$1,488
6	11.00	5.36	711	0.67	12.26	0.043	\$270,343	\$6,940	0.68	12.29	0.040	\$319,907	\$5,000	-\$1,440
7	6.60	5.90	682	0.55	10.96	0.041	\$261,431	\$6,865	0.64	12.21	0.044	\$306,004	\$5,500	-\$1,365
8	15.01	5.48	609	0.63	12.51	0.038	\$287,369	\$7,866	0.70	12.79	0.032	\$364,041	\$5,500	-\$2,396
9	4.00	5.80	790	0.49	10.05	0.055	\$223,008	\$6,503	0.57	10.93	0.056	\$246,997	\$5,500	-\$1,003
10	10.70	5.20	664	0.60	11.87	0.042	\$268,237	\$7,803	0.68	12.57	0.033	\$346,117	\$5,500	-\$2,302
11	5.37	5.81	682	0.58	12.19	0.046	\$265,453	\$7,515	0.66	12.48	0.037	\$326,957	\$5,500	-\$2,015
Average	7.90	5.65	708	0.58	11.54	0.045	\$267,773	\$7,242	0.66	12.18	0.041	\$320,696	\$5,500	-\$1,780

Notes. PF: Probability of patient death one-year post-transplant. The set expected costs are chosen by the social planner to the fixed value of \$5500 to offset the benefit

Table 4 Comparison of over-intense and insufficiently intense citation policy

Data	Social		Provider		Citation Policy		Provider's Performance			
	Offer Rate	h	k_s^*	Utility	k_t	Utility	Optimal Expected Cost	Imposed Expected Cost	Expected Cost	Δ
Over-intense	7.48	2.38	0.47	6.67	0.72	6.69	\$3,081	\$5,500	\$2,420	
Insufficiently intense	7.89	5.97	0.72	13.95	0.41	13.95	\$12,100	\$5,500	-\$6,600	

Notes, SL: social loss, Def: deficiency

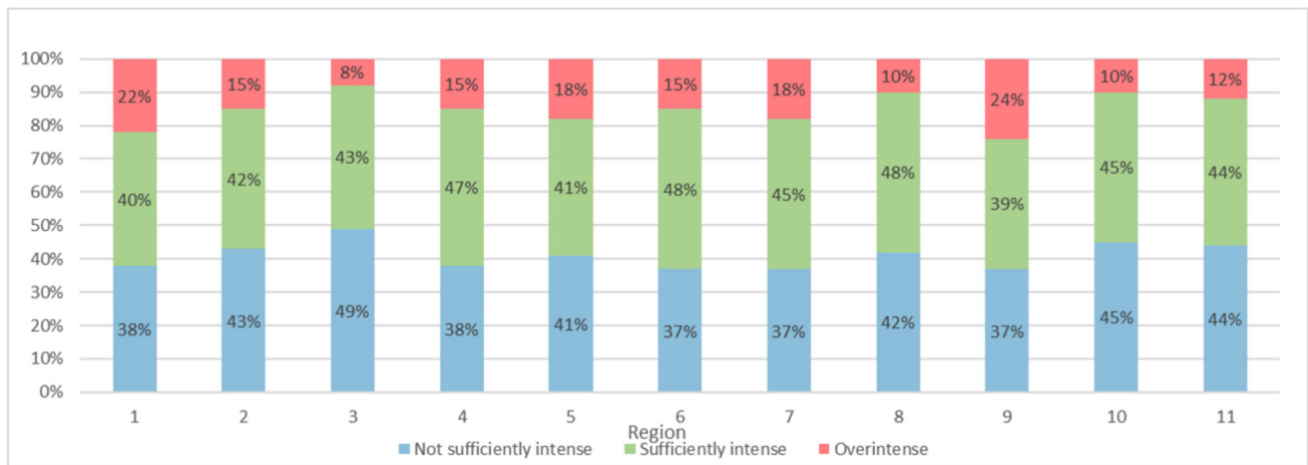


Fig. 3 CoP efficiency across 11 regions. Given the recipients' health profiles and kidney supply demand in each region, on average, the percentages of patients who received both optimal and sub-optimal kidneys due to the over-intense and insufficiently intense oversight policy are demonstrated across all 11 regions

and demand factors determining kidney allocation), decisions leading to outcomes (e.g., transplant choices made by providers in expectation of outcomes), and incentive structures imposed by the quality oversight (e.g., transplant centers' kidney selection and patient acceptance decisions to counter quality oversight). We analytically examine the effect of quality oversight in the form of a citation policy (CP), i.e., citing underperforming providers on social welfare. The efficiency (the ratio of expected utilities under the provider's decision to the socially optimal decision) and equity (a quantity that takes into account different demand-supply ratios among different regions) of the CP are highly correlated with designing well-established risk-adjusted models. To improve the effectiveness and fairness of this policy, we recognize the fundamental components required to be reflected in a risk-adjusted model. As such, we show that as the kidney allocation rules change, an update in the CP is needed to maintain its efficiency. We propose a unique closed-form solution to accept organs. Our model considers several key factors: 1) heterogeneity for both kidney quality and candidate health, 2) organ supply, and 3) the role of decision-makers' risk preferences in accepting offered organs.

The model balances the trade-off between the benefit and the cost of kidney transplantation. The benefit is measured as life-year gains from receiving kidney transplantation which depends on the quality of the accepted donor kidney. The cost includes the candidate's health deterioration as she waits for a better offer. A socially optimal kidney transplantation decision depends on the candidate's health and local donor kidney supply, which maximizes the system's welfare utility function. The optimal CP needs to be designed consistent with key parties' incentives with the goal of minimizing the social

loss resulting from the misalignment between transplant centers' and social planner's objectives and goals. For practical purposes, by employing historical aggregated data, we propose admission policies to admit patients to the waitlist considering the patients' health, the current state of the system, and the kidney supply rates.

Our empirical study presents several important insights regarding effective and efficient kidney transplant decisions and quality oversight. First, we observe from our data that the candidate's blood type and region influence wait time significantly. However, the quality of accepted kidneys is almost the same among recipients with different blood types in different regions. This observation suggests that the role of the candidate's offer rate, which is the reflection of donor kidney supply, is currently widely neglected in making the best transplant decisions. The empirical results based on our model indicate that the factors that influence a candidate's wait time, such as blood type and region supply, should be considered in making the optimal kidney transplant decisions. Furthermore, our empirical study reveals that under the current CoP, the sub-optimal kidney acceptance decisions have been made for 15% and 41% of recipients as a result of over-intense and insufficiently intense citation policy, respectively. Also, close to the optimal decisions have been made only for 44% of recipients. It should be noted that the current regulations for the evaluation of transplantation centers' outcomes are risk-adjusted only based on recipient and donor risk factors.

Our model suggests that quality oversight might be more effective if the CPs take the organ availability for the eleven regions and the different blood types into account. This observation might reduce the tendency of risk-averse behavior

among decision-makers which currently leads to sub-optimal results. The other advantage of supply-based adjustment is that it can be performed under any kidney assignment rule by assessing the effects on individuals' supply or offer rate. For instance, the post-KAS benefits the top 20% of patients by providing higher access to the kidneys, whereas it has the opposite effect on the remaining candidates. However, the regulators still apply the same evaluation criteria in assessing transplant outcomes as before (i.e., the CP remained the same after the introduction of KAS). Not adjusting for the fact that the new kidney assignment causes changes in individuals' offer rate might motivate transplant centers to limit access to care for high-risk (i.e., high-EPTS) patients. The main reason is that high-risk patients do not have access to high-quality kidneys, so their post-transplant outcome could be worse. Not reflecting this concern in the performance assessment for oversight could encourage transplant centers to delay transplantation for these patients leading to potentially increased waitlist mortality. Moreover, the centers may even become more likely to delist high-risk candidates from the waitlist, which in turn may exacerbate access issues.

We also show that risk-seeking and risk-averse behaviors result in sub-optimal outcomes. Our empirical study demonstrates that after CoP, 6 regions out of 11 have adjusted their risk preference class to that reflect more risk neutrality. The remaining 5 regions have become more of a risk seeker type (remaining longer on the waitlist), which could support the assumption of being selective for accepting offered donor kidneys as a result of the oversight policy. Even though our data analysis indicates that 56% of recipients are impacted by receiving sub-optimal kidneys, in general, the social loss and deficiency have decreased across all 11 regions. We also observe an increase in the degree of benevolence among 11 regions. Overall, based on our analysis, a total of 37,064 life years have been saved across all recipients who received donor kidney transplantation after the introduction of CoP in 2007.

Our work leaves some open questions for future research. In addressing our research questions, we develop a stylized model of a single queue for compatible types of patients and kidneys at a local level. For model tractability, we follow the prior literature in making assumptions such as time-homogeneous Poisson arrivals of kidneys and patients. The model does not consider the detailed organ allocation and matching process involving a cross-region network. It also does not consider how decisions by other waitlisted patients impact a focal patient's organ offer and acceptance rate. Future work can investigate the existence and feasibility of a tractable model that better captures these details on organ allocation rules, patient inter-dependencies, competitions

among candidates and providers, as well as multiple listing options.

Appendix A: Summary of notation

Table 5 Definition of rates

μ	Kidney arrival rate
$\mu_{o:i}$	Kidney offer rate
$\mu_{a:i}$	Kidney acceptance rate
μ_l^a	Kidney acceptance rate post-KAS - low-risk
μ_h^a	Kidney acceptance rate post-KAS - high-risk

Table 6 Definition of kidney quality

q	Kidney quality $\in [0, 1]$
$k_{c:i}^*$	Candidate's optimal kidney quality threshold
$k_{p:i}^*$	Provider's optimal kidney quality threshold
$k_{s:i}^*$	Social planner's optimal kidney quality threshold
k_t	General provider's optimal kidney quality threshold
k_l^*	Low-risk candidate's optimal kidney quality threshold post-KAS.
k_h^*	High-risk candidate's optimal kidney quality threshold post-KAS.

Table 7 Definition of utility and expected utility functions

$U_i^c(h_i, q, w_{t,i})$	Candidate's post-transplant utility
$E(U_i^c)$	Candidate's expected post-transplant utility, social welfare utility
$E(U_i^p)$	Provider's expected post-transplant utility
$E(U_l^c)$	Low-risk candidates's expected post-transplant utility post-KAS
$E(U_h^c)$	High-risk candidates's expected post-transplant utility post-KAS

Table 8 Definition of survival and hazard functions

$S_i(k_i, h_i, t)$	One-year post-transplant survival
$F_i(k_i, h_i, t)$	One-year post-transplant failure
$\lambda(q, k, h_i)$	Hazard function post-transplant period
$\lambda_h'(i)$	Post-transplant hazard rate

Table 9 Other definitions

$\eta_{c:i}$	Candidate's risk preference $\in [0, 1]$
δ	Provider's degree of benevolence
$w_{t:i}$	Maximum wait time until receiving an acceptable offer
B^p	Provider's benefit from transplant
B'	Providers's benefit from enlisting candidate
C'	Cost of losing candidates on the waitlist
C_p	Providers's cost of being cited
f_p	Providers's probability of being cited
N	Number of active candidates
N_l, N_h	Number of active low- and high-risk candidates

Appendix B: Proofs

B.1 Proof of Eq. 1

Since $w_{t:i}$ has an exponential distribution with parameter $\mu_{a:i}$, we have:

$$\eta_{c:i} = p(W < w_{t:i}) = 1 - \exp(-\mu_{a:i} w_{t:i}).$$

after rearranging:

$$1 - \eta_{c:i} = \exp(-\mu_{a:i} w_{t:i}) \implies \ln(1 - \eta_{c:i}) = -\mu_{a:i} w_{t:i}.$$

which results in:

$$w_{t:i} = \frac{\ln(\frac{1}{1-\eta_{c:i}})}{\mu_{a:i}}.$$

B.2 Proof of Proposition 1

The kidney quality scores are scaled such that the distribution of Q is uniform, resulting in $P(Q \geq k_{c:i}) = 1 - k_{c:i}$. This is a common assumption in the literature since the Kidney Donor Profile Index (KDPI) score, which reflects kidney quality, is by construction close to uniformly distributed [38].

Substituting Eqs. 7-8 into Eq. 4 results in:

$$U_i^c(h_i, q, k_{c:i}) = m_{h_i} q \left(1 - \frac{\ln(\frac{1}{1-\eta_{c:i}})}{\mu_{o:i} h_i (1 - k_{c:i})} \right).$$

By taking the expected value given $Q \sim U(0, 1)$, we have:

$$E(U_i^c) = \frac{m_{h_i} (1 + k_{c:i})}{2} \left(1 - \frac{\ln(\frac{1}{1-\eta_{c:i}})}{\mu_{o:i} h_i (1 - k_{c:i})} \right).$$

Then, taking the first order derivative results in:

$$\frac{\partial E(U_i^c)}{\partial k_{c:i}} = \frac{m_{h_i}}{2} \left(1 - \frac{2 \ln(\frac{1}{1-\eta_{c:i}})}{\mu_{o:i} h_i (1 - k_{c:i})^2} \right).$$

Setting the above equation to zero results Eq. 16. The second derivative is obtained as:

$$\frac{\partial^2 E(U_i^c)}{\partial k_{c:i}^2} = \frac{m_{h_i}}{2} \left(-\frac{\ln(\frac{1}{1-\eta_{c:i}})}{\mu_{o:i} h_i (1 - k_{c:i})^2} \right).$$

which is always negative, therefore, $k_{c,i}^*$ is the maximizer of $E(U_i^c)$, with the corresponding maximum value given by $E(U_i^c)(k_{c,i}^*)$.

B.3 Proof of Lemma 4.1

Since our model does not consider the effect of competition between candidates, social welfare is

$$S = \sum_{i=1}^N p_i E[U_i^c],$$

and it is easy to see that the sum is maximized when each individual candidate maximizes her own post-transplant expected utility. By substituting $\eta_{c:i} = 1 - \frac{1}{e}$ in in Eq. 11, $k_{s:i}^*$ can be obtained.

B.4 Proof of Proposition 3

From Eq. 20 we get $\exp(-\lambda'(h_i)(1 - k_{p:i})) \geq 1 - \frac{B^p}{C_p f_p}$. Taking the logarithm from both sides results in $-\lambda'(h_i)(1 - k_{p:i}) \geq \ln(1 - \frac{B^p}{C_p f_p})$ or $k_{p:i|CP} \geq 1 + \frac{1}{\lambda'(h_i)} \ln(1 - \frac{B^p}{C_p f_p})$. Thus, the optimal kidney quality threshold under the citation policy consideration given $B^p < C_p f_p$ is obtained as Eq. 21.

B.5 Proof of Proposition 4

Substituting Eqs. 16 and 21 into $k_{p|CP}^* = k_s^*$ results in:

$$1 - \sqrt{\frac{2}{\mu_{o:i} h_i}} = 1 + \frac{1}{\lambda'(h_i)} \ln(1 - \frac{B^p}{C_p f_p}).$$

By simple math calculation and rearrangement we get:

$$-\lambda'(h_i) \sqrt{\frac{2}{\mu_{o:i} h_i}} = \ln(1 - \frac{B^p}{C_p f_p})$$

$$\exp(-\lambda'(h_i) \sqrt{\frac{2}{\mu_{o:i} h_i}}) = 1 - \frac{B^p}{C_p f_p}$$

$$[C_p f_p]^* = \frac{B^p}{\exp(-\lambda'(h_i) \sqrt{\frac{2}{\mu_{o:i} h_i}})}.$$

B.6 Proof of Proposition 5

$Q_{\geq \pi}$ and $Q_{< \pi}$ represent random variables obtained by truncating Q above and below π , respectively. Since Q has a uniform distribution, and $N = N_l + N_h$ we get a thinned Poisson process with rate $\mu_l^a = \frac{\mu}{N}(1 + \pi - k_l)$. The expected wait time until an acceptable kidney offer under the new KAS is $w_l(k_l) = \frac{1}{\mu_l^a}$. The total post-transplant expected utility for a low-risk candidate using the new KAS kidney assignment consists of a component for the low-risk kidneys that only low-risk candidates have access to and the high-risk kidneys with acceptable quality which are shared with all candidates.

$$E[U_l^c] = P(Q < \pi) \frac{\int_{k_l}^{\pi} U_l^c(h_i, q, k) f(q) dq}{\int_{k_l}^{\pi} f(q) dq} + P(Q \geq \pi) \frac{\int_{\pi}^1 U_l^c(h_i, q, k) f(q) dq}{\int_{\pi}^1 f(q) dq}$$

Similar to proof of Eq. 16, by substituting μ_l^a and $w_l(k_l)$ in $E[U_l^c]$ and taking a derivative with respect to k_l Eq. 24 can be obtained.

B.7 Proof of Corollary 6.1

By substituting $\pi = 0.8$ in Eq. 24 and subtracting Eq. 16 from it, one can see $1.8 - \frac{1.61}{\sqrt{\mu_{o,i} h_i}} - 1 + \frac{1.41}{\sqrt{\mu_{o,i} h_i}}$ is always positive as long as $\mu_{o,i} h_i > 0.06$. This requirement is always met by low-risk candidates, meaning that they increase their post-KAS kidney quality threshold.

B.8 Proof of Proposition 6

Similar to proof of Eq. 24 by substituting μ_h^a and the expected wait time of $w_h(k_h) = \frac{1}{\mu_h^a}$ in the post-transplant expected utility function

$$E(U_h^c) = \frac{\int_{k_h}^{\pi} U_h(h_i, q, k) f(q) dq}{\int_{k_h}^{\pi} f(q) dq}$$

Equation 26 can be achieved.

B.9 Proof of Proposition 8

It is trivial for any queue size $l \leq N$, so we present proof for $l > N$. From Eq. 31, the service provider should accept an arriving patient if $S(h', l, \mu) \geq \frac{C'}{B' + C'}$. For given queue length l and kidney supply rate μ , $S(h', l, \mu)$ is increasing in h' , so the optimal admission policy is the single threshold policy. Let $h'_a(l, \mu)$ be such that: $S(h'_a(l, \mu), l, \mu) = \frac{C'}{B' + C'}$, the optimal policy is to accept a patient if $h' \geq h'_a(l, \mu)$.

For a given kidney supply rate μ , from Eq. 30, we can show that the waiting time $W'(l, \mu)$ is stochastically increasing in the queue size l , that is if $N < l_1 < l_2$ then $W'(l_1, \mu) <_{st} W'(l_2, \mu)$. Note that the waiting time $W_t(\mu)$ is independent of the queue size l if $l \geq N$. Hence,

$$S(h', l_1, \mu) = Pr(W'(l_1, \mu) + W_t(\mu) \leq h') > S(h', l_2, \mu) = Pr(W'(l_2, \mu) + W_t(\mu) \leq h')$$

This shows that $h'_a(l_1, \mu) < h'_a(l_2, \mu)$.

On the other hand, for a fixed value of l , from equation (.), we have $T_m(\mu)$ is stochastically decreasing in the kidney supply rate μ , i.e., for $\mu_1 < \mu_2$, we have: $T_m(\mu_2) <_{st} T_m(\mu_1)$. Thus,

$$W'(l, \mu_2) = \sum_{l_c}^l T_m(\mu_2) <_{st} W'(l, \mu_1) = \sum_{l_c}^l T_m(\mu_1)$$

Moreover, $W_t(\mu_2) <_{st} W_t(\mu_1)$ and hence, $S(h', l, \mu_2) = Pr(W'(l, \mu_2) + W_t(\mu_2) \leq h') > S(h', l, \mu_1) = Pr(W'(l, \mu_1) + W_t(\mu_1) \leq h')$. This indicates that: $h'_a(l, \mu_2) < h'_a(l, \mu_1)$, i.e. the optima health threshold is decreasing in the kidney supply rate.

Appendix C: Non-linear utility functions and stochastic gradient descent

Real-world scenarios often exhibit diminishing marginal returns, where the incremental benefit of receiving a higher-quality kidney decreases as quality improves. To capture this, we model the benefit function $B(h_i, q)$ as an exponential form:

$$B(h_i, q) = m_{h_i} (1 - e^{-\lambda_q q}),$$

where m_{h_i} represents the maximum potential benefit for a candidate with health level h_i , and λ_q controls the curvature of the benefit with respect to kidney quality (q). The cost function $C(h_i, w_{t,i})$, which represents health deterioration over wait time $w_{t,i}$, is modeled logarithmically:

$$C(h_i, w_{t,i}) = \log\left(1 + \frac{w_{t,i}}{h_i}\right).$$

This ensures a slow initial increase in costs that accelerates as wait time grows and higher health levels result in a slower increase in cost, reflecting that healthier candidates experience less impact from the same wait time. While the linear utility model allows for closed-form solutions, the non-linear model introduces complexities that require a numerical approach. We adopt Stochastic Gradient Decent (SGD) to

Table 10 Outcomes under non-linear utility functions for varying health levels

Health (h_i)	Optimal Threshold ($k_{c;i}^*$)	Expected Utility
1	0.45	3.93
3	0.61	8.97
5	0.66	13.08
7	0.72	15.29

optimize the acceptance threshold $k_{c;i}^*$ by maximizing the candidate's expected utility:

$$\mathbb{E}[U_i^c] = \frac{\int_{k_{c;i}}^1 U_i^c(h_i, q, w_{t;i}) f(q) dq}{\int_{k_{c;i}}^1 f(q) dq}.$$

The gradient of the expected utility with respect to $k_{c;i}$ was approximated numerically:

$$\frac{\partial \mathbb{E}[U_i^c]}{\partial k_{c;i}} \approx \frac{\mathbb{E}[U_i^c(k_{c;i} + \epsilon)] - \mathbb{E}[U_i^c(k_{c;i} - \epsilon)]}{2\epsilon},$$

where ϵ is a small step size. The threshold was updated iteratively using:

$$k_{c;i}^{(t+1)} = k_{c;i}^{(t)} - \eta \frac{\partial \mathbb{E}[U_i^c]}{\partial k_{c;i}},$$

with η as the learning rate. The results from the simulation using the non-linear model highlight its ability to capture the complexities of transplantation, including diminishing returns and accelerating costs. To evaluate the model, we perform a simulation using SGD to optimize the acceptance threshold $k_{c;i}^*$. The results for some pre-determined health levels are summarized in Table 10, showcasing the optimal thresholds and expected utilities across different health levels.

Appendix D: Sensitivity analysis, parameter estimation, and discard variations

D.1 Organ offer probability

In reality, the probability of receiving an offer may increase as the patient waits longer. While relaxing this implicit assumption could broaden the parameter space for optimal policies, it would not qualitatively alter our results.

Our numerical analysis shows that variations in organ offer probabilities can influence decision thresholds and expected utility; however, the impact is most significant at lower organ offer probabilities and for sicker patients.

For instance, as shown in Figs. 4, 5, and 6, an increase in the probability of an organ offer from 15% to 25% results in an average increase in the optimal kidney quality threshold by 0.12, 0.11, and 0.08 for the health levels of 1, 5, and 10, respectively. For the expected utility, we observe average increases of 1.14, 1.15, and 1.21 years for the same health levels. When the probability of receiving an offer is greater than 0.25, the improvement in the decision threshold is less than 0.10. When the offer probability exceeds 0.55, the improvement in acceptance decision is less than 0.05. The improvement in post-transplant utility is less than one expected post-transplant year when the offer probability surpasses 0.25.

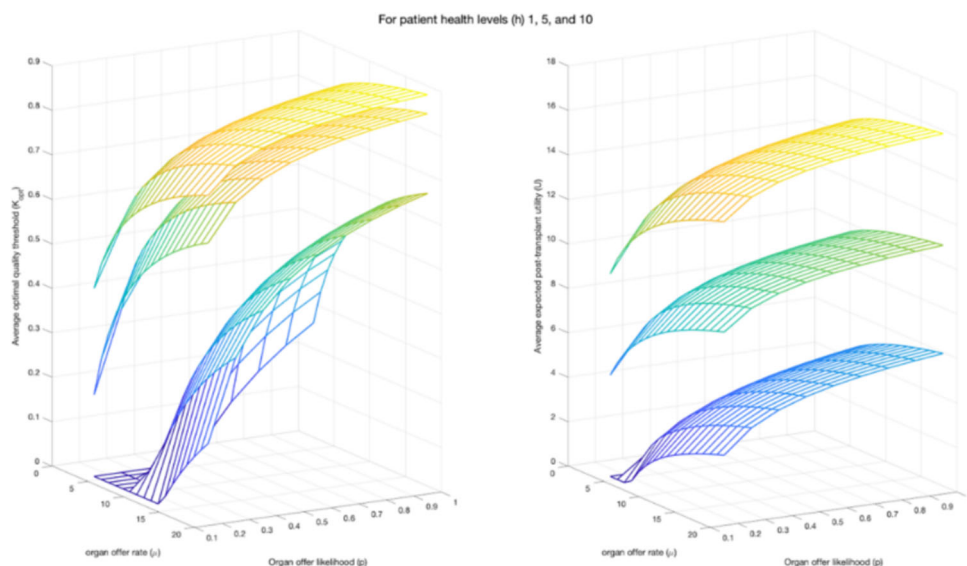
At higher probabilities, such as an increase from 65% or above, the associated improvement in the optimal threshold is less than 0.03, and the increase in expected utility is less than 0.4 years. These modest changes apply across all health levels.

D.2 Parameter estimation

To report the observed values in Table 1, we calculate the average KDPI, k_t , for all transplants performed in the US before and after 2007. We divide the transplants into two groups based on whether they are performed before or after the introduction of the CoP in 2007. To calculate the post-transplant utility, U_i^c , for each group, we use Eq. 4. To find the value of k_s^* , we use Eq. 16, and compute the corresponding utility for each transplantation across all regions. We then took the average of all k_t , k_s^* , and utilities and reported the values for each of the 11 regions. To obtain or estimate the parameters required for Eqs. 4 and 16 we do the following:

1. For each individual patient, the SRTR data set includes the wait time till transplant, and we used that information for computing $w_{t;i}$
2. We use SRTR data to calculate the organ offer rate. We estimate the organ supply by analyzing all viable organs that are procured for transplant purposes. We then adjust the offer rate for each patient by considering their blood type and the number of imported or exported organs to and from each region. To adjust the offer rate based on a patient's blood type, we examine the probability of them receiving kidneys from blood types A, B, AB or O and adjust their offer rate accordingly. For instance, if the patient has blood type AB, we adjust their offer rate based on the likelihood of receiving kidneys from blood types A, B, AB, or O.
3. We utilize data from USRDS and SRTR to derive life expectancy estimates for dialysis patients awaiting transplantation, segmented across various age brackets. This is achieved by dynamically calculating the queue size for

Fig. 4 Kidney acceptance decision threshold and the expected post-transplant utility across different values of organ offer rate and organ offer likelihood for patient health level of 1, 5, and 10



each bracket, taking into account both patient arrivals to and departures from the dialysis cohort, thus reflecting the cohort's fluid nature. Adjusted death rates, sourced from the USRDS and expressed per 1000 person-years, provided a measure of mortality risk within each age group. To estimate the 'Exposure to Risk' for patients within each age bracket, we innovated by including not only the existing cohort size but also integrating a measure for new arrivals - presuming these individuals contribute to half the period's risk exposure. This adjustment allows for a more accurate representation of the risk faced by individuals in the queue. Subsequently, patient health expectancy estimates are derived through the inversion of this risk-adjusted death rate. By aggregating and averaging these health expectancy figures across

all brackets and regions, we establish a comprehensive overview of overall life expectancy for the dialysis patient population, factoring in both the continual introduction of new patients and patient mortality. In the following, we outline the specific steps undertaken in our calculation, emphasizing the systematic integration of data and the application of risk-adjusted metrics to accurately assess life expectancy.

- (a) To calculate the queue size for each age bracket, defined generically as $[a, b], [c, d], [e, f]$, etc., we dynamically calculated the queue size. This calculation took into account both patient arrivals $A_{[a,b]}, A_{[c,d]}, A_{[e,f]}$, etc and departures $D_{[a,b]}, D_{[c,d]}, D_{[e,f]}$, etc from the dialysis cohort, reflecting the cohort's fluid nature.

Fig. 5 Changes in Kidney transplant decision across different offer probabilities for patient health levels of 1, 5, and 10

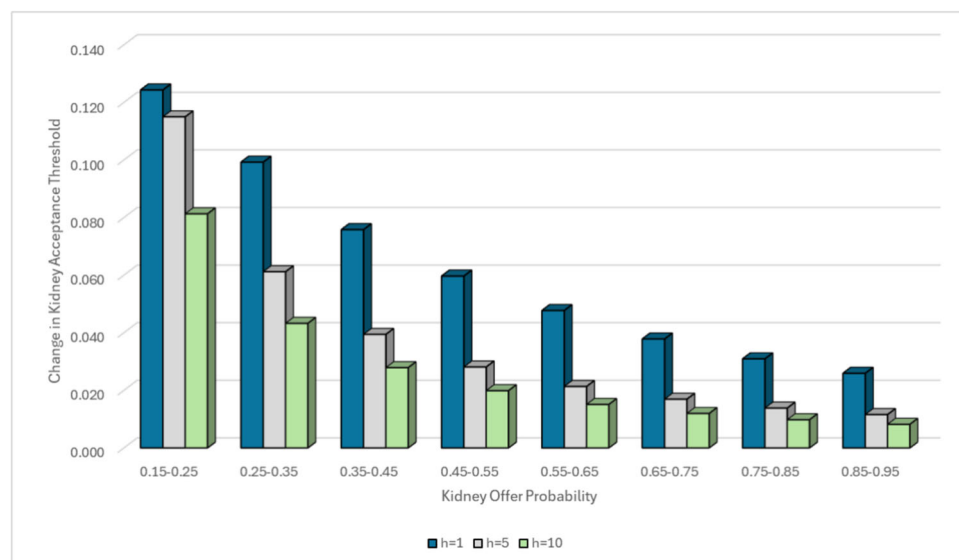
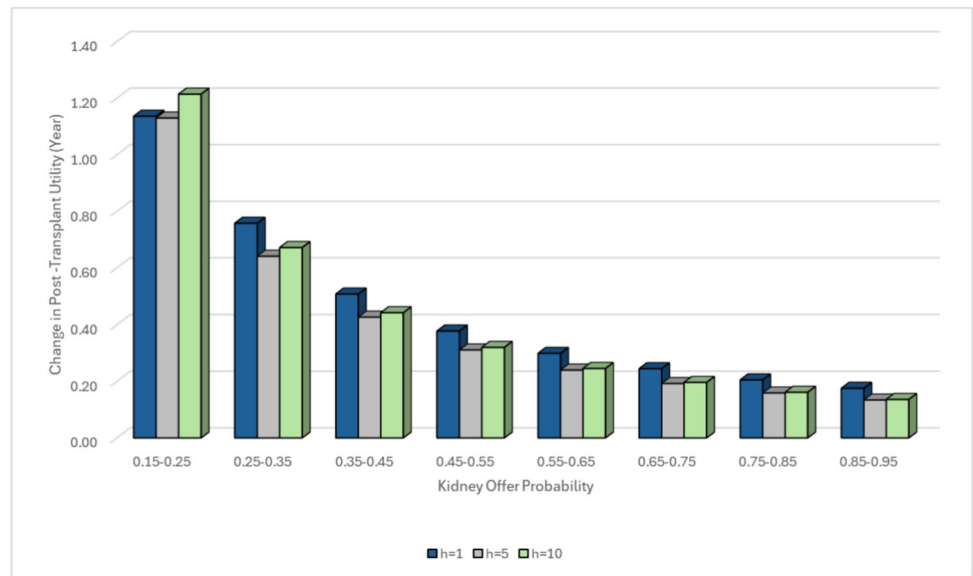


Fig. 6 Changes in Kidney expected post-transplant utility across different offer probabilities for patient health levels of 1, 5, and 10



- (b) We obtain adjusted death rates $DR_{[a,b]}$, $DR_{[c,d]}$, $DR_{[e,f]}$ sourced from the USRDS and expressed per 1000 person-years, to provide a measure of mortality risk within each age group.
- (c) To estimate the 'Exposure to Risk' $E_{[a,b]}$, $E_{[c,d]}$, $E_{[e,f]}$, etc for patients within each age bracket, we accounted for both the existing cohort size and integrated a measure for new arrivals, assuming these individuals contribute to half the period's risk exposure. This was calculated as follows: $E_{[x,y]} = Q_{[x,y]} + 0.5A_{[x,y]}$ where $[x, y]$ represents any age bracket such as $[a, b]$, $[c, d]$, or $[e, f]$.
- (d) Patient's health expectancy estimate for each bracket was then derived through the inversion of the risk-adjusted death rate, calculated as: $HealthExpectancy_{[x,y]} = \frac{1}{DR_{[x,y]}/1000}$
- (e) By aggregating and averaging these life expectancy figures across all brackets and regions; we computed overall life expectancy for the dialysis patient population.

- (iii) The observed degree of benevolence δ can be calculated by solving the following equation:

$$k_{t:i} = (1 - \delta)k_{p:i}^* + \delta k_{s:i}^* \quad (D1)$$

By definition, the threshold for a purely self-interested provider, $k_{p:i}^*$, is equal to $\underline{k} = 0$ which results in $\delta = \frac{k_{t:i}}{k_{s:i}^*}$.

To calculate the optimal CP, we use $\lambda'(h_i)$, μ_o , and h_i (estimated from the data) in Eq. 22 and calculate $[C_p f_p]^*$. The current CP, $C_p f_p$, can be estimated from the data by substituting the probability of failure $F(k_{s:i}^*, h_i, t)$ with $F(k_{t:i}, h_i, t)$, the probability of failure given the kidney quality threshold $k_{t:i}$. We introduce Δ as the difference between the provider's expected cost and the optimal-citation expected cost given its decision $k_{t:i}$ as

$$\Delta = (C_p f_p - [C_p f_p]^*) F(k_{t:i}, h_i, t). \quad (D2)$$

D.3 Formulas used for the empirical analysis

We measure social loss, deficiency, and degree of benevolence for each candidate as follows.

- (i) The social loss attributable to a candidate i is defined as the difference between the social welfare utility function's value given a social optimal kidney transplant decision ($k_{s:i}^*$), and the provider's decision observed from data ($k_{t:i}$).
- (ii) We measure *efficiency* as the expected utility under the provider's decision, $E(U_i^c | k_{t:i})$, relative to the expected utility under the socially optimal decision, $E(U_i^c | k_{s:i}^*)$, and define *deficiency* as $1 - \text{efficiency}$.

If Δ is close to zero, then the current $C_p f_p$ is the optimal CP. In other words, the provider's perception of CP leads to setting its threshold k_t equal to the social optimum k_s^* . Table 3 presents the current and the optimal CP across 11 regions in the U.S. We compute Δ for every recipient across 11 regions after the implementation of CoP.

In this section, we analyze the effect of the citation policy on kidney acceptance decisions. We estimate the current citation policy, $C_p f_p$, using providers' data to assess whether the current citation policy is close to the optimal policy, $[C_p f_p]^*$. In case the current performance assessment is not optimal, we measure how far it is from the optimal citation policy and how much it needs to change to reach the optimal goal.

Table 11 Kidney Discard Rate Before and After Introduction of CoP

Region	Kidney Discard Rate Before CoP	Kidney Discard Rate After CoP	% Change in Kidney Discard
1	11.8%	19.0%	61.0%
2	15.3%	24.5%	59.9%
3	14.0%	18.0%	28.6%
4	12.0%	19.0%	58.3%
5	11.0%	19.0%	72.7%
6	11.5%	17.0%	47.8%
7	11.5%	20.0%	73.8%
8	13.5%	17.5%	29.6%
9	12.0%	21.0%	75.0%
10	14.2%	20.5%	44.4%
11	13.1%	19.0%	45.0%

D.4 Insights into the organ discard variations across regions

Do findings on regional variations in the intensity of oversight policies align with organ discard rate trends across regions? To compare, we tabulate organ discard rates before and after CoP implementation. Table 11 summarizes this information. The analysis reveals significant regional variations in the percentage change in kidney discard rates. Regions 9, 7, 5, and 1 show the highest increases in discard rates, while Regions 3 and 8 have the lowest changes. These findings align with insights from Fig. 3, categorizing regions by oversight policy intensity-insufficient, sufficient, and over-intense-and showing the distribution of patients receiving both optimal and sub-optimal kidneys. Based on the figure, regions with the highest increases in discard rates (Regions 9, 7, 5, and 1 as shown in Table 11) exhibit the highest proportions of responses to over-intense oversight (as shown in Fig. 3), suggesting a correlation between stringent CoP criteria and increased organ discard rates. Conversely, Regions 3 and 8, with the lowest percentage changes, demonstrate relatively balanced oversight intensity. The comparison highlights how regional differences in CoP implementation correlate with organ discard rates, supporting the need for a customized approach to CoP criteria that considers regional health profiles and kidney supply-demand dynamics.

Author Contributions

- Conceptualization: Zahra Gharibi, Mehmet Ayyaci, Michael Hahsler.
- Methodology, Formal analysis and investigation Writing - original draft preparation, Writing - review and editing: Zahra Gharibi, Michael Hahsler, Mehmet Ayyaci, Hung Do.

Data availability statement Ethics approval to use STAR File datasets was obtained from OPTN.

Declarations

Conflicts of Interest The authors declare no conflict of interest.

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